Design of multilayer vascular grafts that reproduce the stress–stretch curve of each artery: material bases, integer laminae, thick-walled tubes and a neural network

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Abstract

A vascular graft whose distensibility differs from that of the artery it replaces promotes thrombosis and intimal hyperplasia. Published design methods match a single number —the compliance of an average artery—. Here we present a method to design, patient by patient, a multilayer graft that reproduces the *complete* curve of the artery, including its J shape. Each graft material is idealized as a fiber that bears no load until it reaches its own stretch (truncated line), and the artery is approximated by a sum of such materials, chosen once for the whole population, with at most 20 laminae per material. The method is first validated on the circumferential stress–circumferential stretch curve, $\sigma_{\theta\theta}(\lambda_\theta)$, which depends neither on pressure nor on wall geometry. With 41 human carotid arteries (Masson et al.), the minimax integer design —the best integer design that exists, by integer linear programming— meets the target (worst error $\leq 3\,\%$ of the range) for all 41 subjects with 6 materials. A single neural network, which chooses the laminae of each material knowing those of the previous ones (chained network) and is trained inside the band of curves of the population, its domain of validity, directly gives the integer laminae in one evaluation, without optimizing: with a ± 1 lamina search it meets the target for the 31 training subjects, the 10 test subjects and 53 of 63 curves from three other studies —common, external and internal carotid arteries, isolated layers, patients with plaque, in vivo measurements—, with other constitutive laws; the minimax meets it for 57 of those 63, and for the rest no integer design reaches the target. The method is then applied to the pressure–diameter curve, for which the graft is a thick-walled tube: designed with Laplace's law, no subject meets the target once manufactured, whereas with the thick-walled model 40–41 of 41 do. Design becomes a nonlinear integer problem, solved in three steps: initialization, nonlinear least squares, and minimax integer search against the fitted curve. There, a network used as a starting point avoids the local minima that, with measured curves, lead to unusable designs (worst error from 468.7 % down to 2.3 %).

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1 Introduction

The failure of small-diameter vascular grafts (thrombosis, intimal hyperplasia) is largely due to mechanical factors: disturbed flow, insufficient strength and compliance mismatch between graft and artery [14]. In the porcine aorta, a graft with a compliance-matching layer preserved pulse pressure, wave velocity and distensibility, which deteriorated when the layer was removed [44]; in rats, a graft whose composition was optimized for the compliance of the host vessel remained matched at four weeks, with less inflammation and more contractile smooth muscle [18]. Matching the mechanics of the artery improves the outcome.

Several studies optimize the design to this end: crosslinking time and thickness of a biopolymer [22], layer proportions and thicknesses of a multilayer graft [49], a nitinol reinforcement [43] or a reinforcing coil [28], and degradable scaffolds that remodel into a neovessel [48, 25]. All of them match *one number* (compliance or distensibility) of an *average* artery. The response of an artery, however, is J-shaped: soft at low stretch, when elastin and smooth muscle carry the load, and stiff at high stretch, when collagen fibers are recruited [23, 27]. A single number does not describe that shape, and the shape changes from patient to patient and from artery to artery.

This work designs the graft so that it reproduces the *complete* curve of *each* artery. The method has two levels, presented in this order:

1. **Material level (validation).** The target curve is the circumferential stress as a function of circumferential stretch, $\sigma_{\theta\theta}(\lambda_\theta)$. It is a property of the tissue: it does not depend on pressure, radius or wall thickness, and is therefore common to any artery whose wall has a J-shaped response. At this level the graft is a sum of materials, the design is linear, and every step of the method can be compared with an exact solution.

2. **Tube level (application).** The target curve is the pressure borne by the wall as a function of diameter. Geometry and the thick-wall model come into play, the design is no longer linear, and the conclusions of the first level are used.

The contributions are: (i) a basis of fiber-like materials chosen once for the whole population; (ii) the optimal integer design, with at most 20 laminae per material, by integer linear programming (minimax); (iii) a single chained neural network that directly gives the integer laminae in one evaluation, validated separately on training subjects, test subjects and curves from three other studies, within a domain of validity set beforehand; (iv) the demonstration that, at the tube level, the design must be carried out with a thick-walled model, which makes it a nonlinear integer problem; and (v) a three-step procedure in which a network acts as a starting point.

2 Data

2.1 Constitutive law and subjects

Masson et al. [35] measured non-invasively, over the cardiac cycle, the diameter and pressure of the common carotid artery of 16 normotensive (NT) and 25 treated hypertensive (HT) subjects, and fitted to each one a model of the intima–media wall with three contributions: neo-Hookean elastin, four families of collagen fibers with exponential response (axial, circumferential and two diagonal at $\pm\alpha$) [23], and smooth-muscle tone [40]. With fixed axial stretch λ_z and incompressibility,

$$\sigma_{\theta\theta} - \sigma_{rr} = c\,(\lambda_\theta^2 - \lambda_r^2) + \lambda_\theta^2 \sum_{k=1}^{4} c_{1,k} E_k\, e^{c_{2,k} E_k^2} \sin^2 \varphi_k + T_m \lambda_\theta \Big[1 - \Big(\frac{\lambda_m - \lambda_\theta}{\lambda_m - \lambda_0}\Big)^2\Big], \tag{1}$$

with $E_k = \lambda_k^2 - 1$, $\lambda_r = 1/(\lambda_\theta \lambda_z)$ and λ_k the stretch of each family (λ_z, λ_θ and $\sqrt{\lambda_\theta^2 \sin^2\alpha + \lambda_z^2 \cos^2\alpha}$). The 14 parameters of each subject are listed in their Tables 2 and 4 (Appendix A). Ten subjects (4 NT and 6 HT) are held out for testing and play no part in any choice made by the method.

2.2 Stress curves

The target curve of the first level is $\sigma_{\theta\theta}(\lambda_\theta)$ at the media–adventitia boundary, at 999 values of λ_θ between 1.02 and 1.16, computed with (1); at that boundary the radial stress equals the perivascular pressure of Masson's model. Recomputed in this way, it reproduces the published mean curves (Figs. 5a and 7a of [35]) with a root-mean-square error of 1.2 to 1.7 % of their mean value, of the order of the digitization error. The 41 curves are J-shaped (monotonic and convex) and range from 4.7 to 49 kPa at $\lambda_\theta = 1.02$ and from 54 to 290 kPa at 1.16 (Fig. 1). The mean curves published by Masson et al. (Fig. 5a: young and older subjects; Fig. 7a: normotensive and hypertensive) are superimposed in the figure and fall on the recomputed curves. Since pressure is also expressed relative to 13.332 kPa (100 mmHg), stress is additionally shown relative to a reference stress of 150 kPa. In dimensionless form, $(\sigma - \sigma_{\min})/\Delta\sigma$, normotensive and hypertensive subjects have almost the same shape; the difference lies in the stress level.

Two bounding curves enclose the 41 subjects: y_1, a curve of Masson's law with parameters within the range of the subjects, which starts at the lowest measured stress and lies below all of them, and y_2, the stiffest subject extended up to 300 kPa. The *working space* consists of 22 500 synthetic curves of Masson's law within that band, with parameters of 2 to 4 training subjects mixed and perturbed, or drawn between the minimum and maximum of the subjects.

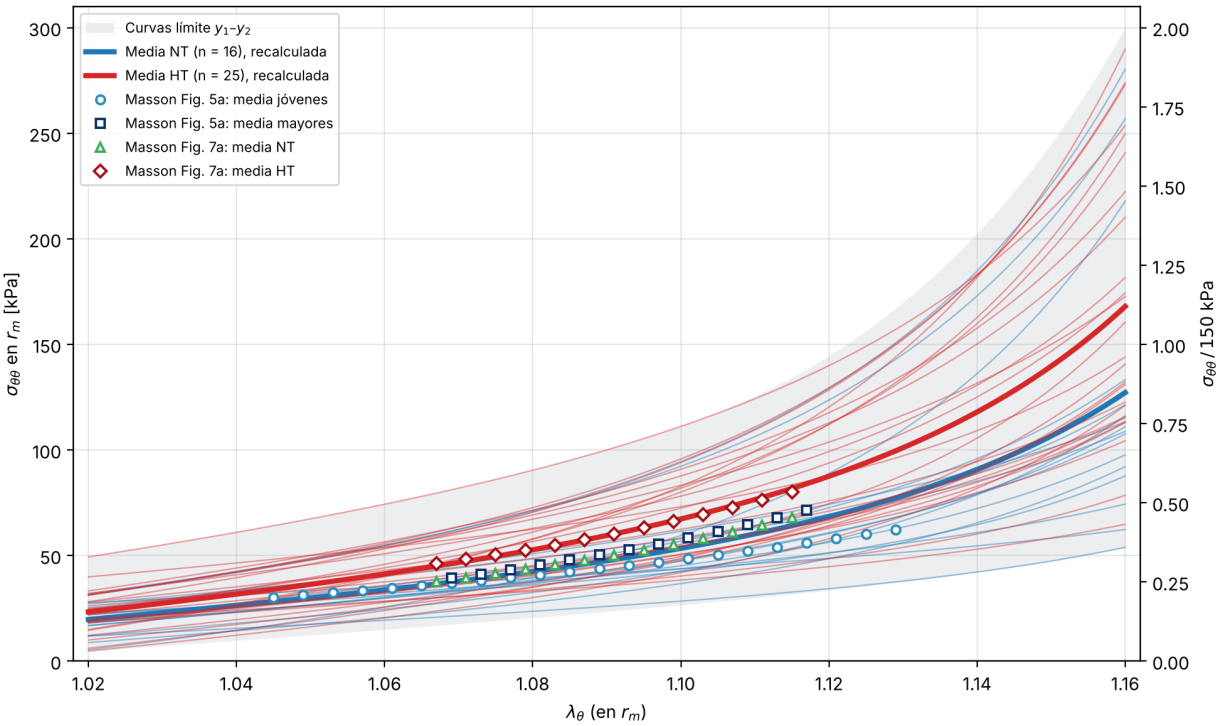


Figure 1: The 41 subjects: circumferential stress at the media–adventitia boundary; left axis in kPa, right axis relative to 150 kPa. Gray band: bounding curves y_1–y_2. Blue: normotensive; red: hypertensive; thick lines: recomputed means of each group. Markers: mean curves digitized from Masson et al. [35], Fig. 5a (young, older) and Fig. 7a (normotensive aged 39 or more, hypertensive); they cover the λ_θ interval shown in the original work.

3 Validation of the method on the stress–stretch curve

3.1 Materials: truncated lines

One lamina of material j contributes

$$A_j(\lambda) = S_j \max(0,\ \lambda - k_j)\,: \tag{2}$$

nothing up to its *knot* k_j, and then a straight line of slope S_j. It idealizes a crimped fiber that bears no load until it is straightened. The graft is the sum $\sigma_{\text{gr}} = \sum_j c_j A_j = A\,c$, where c_j is the number of laminae of each material and $A \in \mathbb{R}^{999\times N}$ is the *basis*. For fixed knots, the problem is linear.

Choice of the knots. The knots are chosen once, for the whole population, by minimizing a function $J(k)$ that combines three requirements on the working space and the training subjects:

- **Accuracy: CVaR 99 %.** For each curve, the maximum pointwise error of the least-squares fit, $e_{\max}$ (Eq. 3), is computed. The CVaR 99 % (conditional value at risk [42]) is the mean of $e_{\max}$ over the 1 % of curves with the largest error. A smaller term, 0.25 times the mean NRMSE, is added so that the typical fit also improves. Minimizing the mean over all curves would leave the worst ones uncontrolled; minimizing only the worst one would make the choice depend on a single curve. The CVaR lies between the two: it addresses the tail of worst curves without depending on a single one.
- **Stability: condition number.** With the columns of A scaled to a maximum of 1, the condition number is $\kappa(A) = s_{\max}/s_{\min}$, the ratio of the largest to the smallest singular value [50]. It measures how much a relative error in the curve can be amplified in the

amounts c: a 1 % error in y can produce up to $\kappa(A)$ % error in c. If two knots are very close, their columns become similar, $s_{\min} \to 0$, and the design becomes unstable (large amounts of opposite sign that cancel each other). We require $\kappa(A) < 30$; the adopted bases have 28.0 (6 materials) and 19.0 (5).

- **Sign.** The amounts of the unconstrained fit, $c = A^{+}y$, must be non-negative (with a tolerance of 0.01 lamina) for the 22 500 curves of the working space, the training subjects and the bounding curves, because a negative amount of material cannot be manufactured. The function J adds a penalty proportional to the negative amounts. Since the sign of c_j does not depend on the slope S_j (which only scales the column), this condition is determined by the knots.

The search is global, by *differential evolution* [47]: a population of candidate knot sets is improved generation after generation; each candidate is perturbed by adding the scaled difference between two others, and the result replaces the candidate if it yields a lower J. It requires no derivatives, which J does not have (the max, the sorting in the CVaR and the penalties make it non-differentiable), and it explores the whole domain, so it does not get trapped in the first local minimum. The best candidate is refined with the *Nelder–Mead* simplex method [37], which is also derivative-free: it moves a polytope of $N + 1$ vertices through reflections, expansions and contractions until it closes in on the minimum.

Constraints on the knots. Knots 2 to N lie within the domain, at least 0.008 apart (if they come together, $\kappa(A)$ grows), and the last one more than 0.012 from the upper end 1.16 (otherwise its material acts on very few points and cannot be distinguished from the others). The first one, k_1, is sought between 0.6 and 0.98, *before* the domain, for a physical reason: at $\lambda_\theta = 1.02$ all arteries already carry stress (4.7 to 49 kPa), and at that point the only material that can supply it is one that is already working. That material plays the role of elastin, which bears load in the unloaded artery. A knot $k_1 < 1$ means that the lamina is mounted prestretched: with $k_1 = 0.82$, at $\lambda = 1$ it is already stretched by $1/0.82 = 1.22$. The upper bound 0.98 ensures that it is active over the whole domain with a margin. The lower bound, 0.6 (prestretch of 1.67), restricts the search to manufacturable values. Neither bound is active at the optimum ($k_1 = 0.82$ with both 5 and 6 materials): the choice does not depend on them.

Slopes. $S_j = \mathrm{SLOPE}_{\max,j}/20$, where $\mathrm{SLOPE}_{\max,j}$ is the largest slope that material j must supply in the working space: thus no curve requires more than 20 laminae of one material. The scale does not change the curve (what matters is $c_j S_j$), but it sets the size of one lamina.

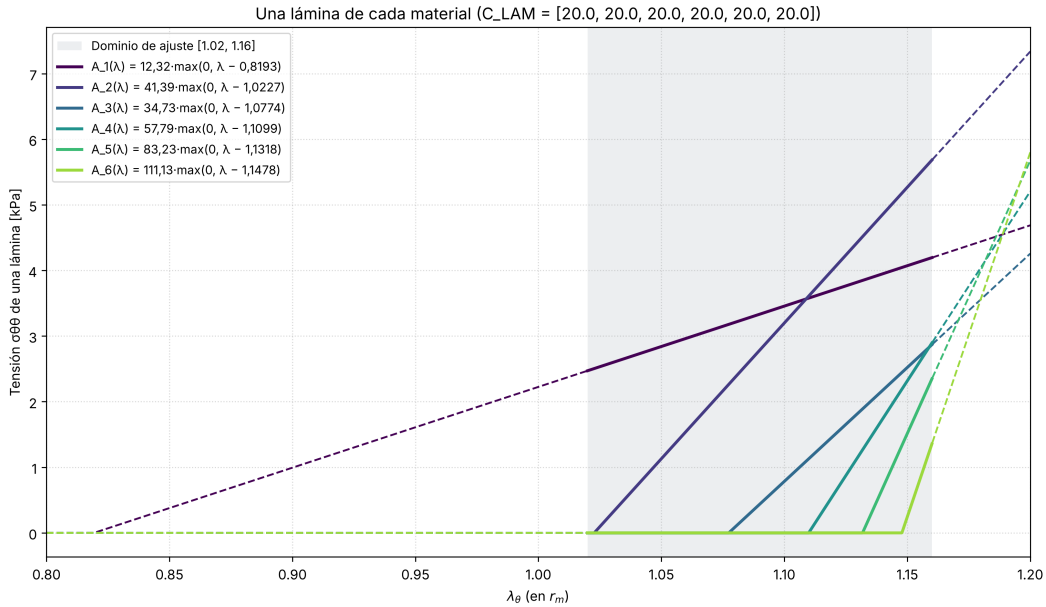


Figure 2: The 6-material basis: one lamina of each. The first knot (0.82) lies before the domain, like elastin, which already works in the unloaded artery; the others cluster where the arterial curve bends most.

Table 1: Adopted bases: knots k_j and slope of one lamina S_j [kPa].

	1	2	3	4	5	6
6 materials, k_j	0.8193	1.0227	1.0774	1.1099	1.1318	1.1478
S_j	12.32	41.39	34.73	57.79	83.23	111.13
5 materials, k_j	0.8228	1.0243	1.0881	1.1226	1.1447	—
S_j	12.56	43.57	51.92	88.74	129.08	—

3.2 Metric

For a target curve y with range $\Delta = \max y - \min y$,

$$e_{\max} = \frac{100}{\Delta} \max_i |y_i - \hat{y}_i| \quad [\%], \qquad E_{\text{rms}} = \frac{100}{\Delta} \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}, \tag{3}$$

where $\hat{y}$ is the graft curve. $e_{\max}$ is the worst point; E_{rms} (or NRMSE), the typical deviation. Dividing by the range makes the criterion equally demanding for a soft and a stiff artery. Target: $e_{\max} \leq 3\,\%$. Every design is evaluated against the original curve.

3.3 Real-valued amounts: least squares

With a fixed basis, the best quadratic fit is $c = A^{+}y$, where A^{+} is the Moore–Penrose pseudoinverse [38], computed by singular value decomposition [19]: exact, unique and instantaneous. With the adopted knots the condition number is 19.0 (5 materials) and 28.0 (6 materials), below the threshold of 30, and no curve of the working space nor any subject requires a negative amount. The fit meets the target in all 41 subjects, with a worst error of 0.88 % (6 materials; median 0.67 %) and 1.41 % (5 materials; median 1.07 %), and in every curve of the working space.

3.4 Integer laminae: the minimax design

A graft is manufactured with an integer number of laminae of each material, $z \in \{0, 1, \ldots, 20\}^N$: at most 20 laminae per material, so as not to complicate manufacturing (with the adopted slopes,

no curve of the working space needs more). The reference integer design does not round any real-valued solution: it directly searches for the integers that leave the narrowest tunnel around the artery curve,

$$\min_{z,\tau} \ \tau \quad \text{s.t.} \quad -\tau \le \frac{100}{\Delta}(y_i - (Az)_i) \le \tau, \qquad z_j \in \{0, 1, \ldots, 20\},$$

a mixed-integer linear program (Chebyshev approximation [11] with integers) solved by branch and bound with HiGHS [26], in about 50 to 65 ms per curve. Its only data are the curve y and the basis A, and what it returns is the *global optimum*: the integer design with the least $e_{\max}$ that exists. If the minimax does not meet the target for a curve, no combination of integer laminae does. It is therefore the reference against which the neural network is measured. Within the domain of validity (Section 3.6) the 20-lamina cap does not bind: without the cap, none of the 63 external curves of Section 3.8 needs more than 18 laminae of one material, and for them and for the 41 subjects the minimax $e_{\max}$ is the same with and without the cap (within 0.01 points).

3.5 The chained neural network

The minimax solves an integer optimization problem for each curve. A neural network, instead, learns once the relation between the curve and its design, and then gives the integer laminae of a new graft in a single evaluation, without optimizing. To make it give integers, each output is a choice among classes: for material j, the probability $p_j(n)$ of using $n = 0, 1, \ldots, 20$ laminae (softmax, $K = 21$ classes), and the network returns the most probable n. A single network, the one in Fig. 3, is used throughout the first level.

Architecture. A shared trunk, $999 \to 256 \to 128 \to 64$ (ReLU [36], layer normalization [5] and L2 regularization of 10^{-5}), summarizes the curve in a vector h of 64 numbers; the input is the standardized $\log(y + 22)$. There is one head per material. Head j receives h and the laminae already chosen, $n_1 \ldots n_{j-1}$ (divided by 20), goes through two layers (128 and 64 neurons, ReLU) and chooses n_j. It is used in order —material 1, then 2 knowing 1, and so on up to the last—, so it can compensate between materials: if it put one lamina too many in one, it can put one fewer in the next. It has 407 614 parameters with 6 materials and 389 033 with 5.

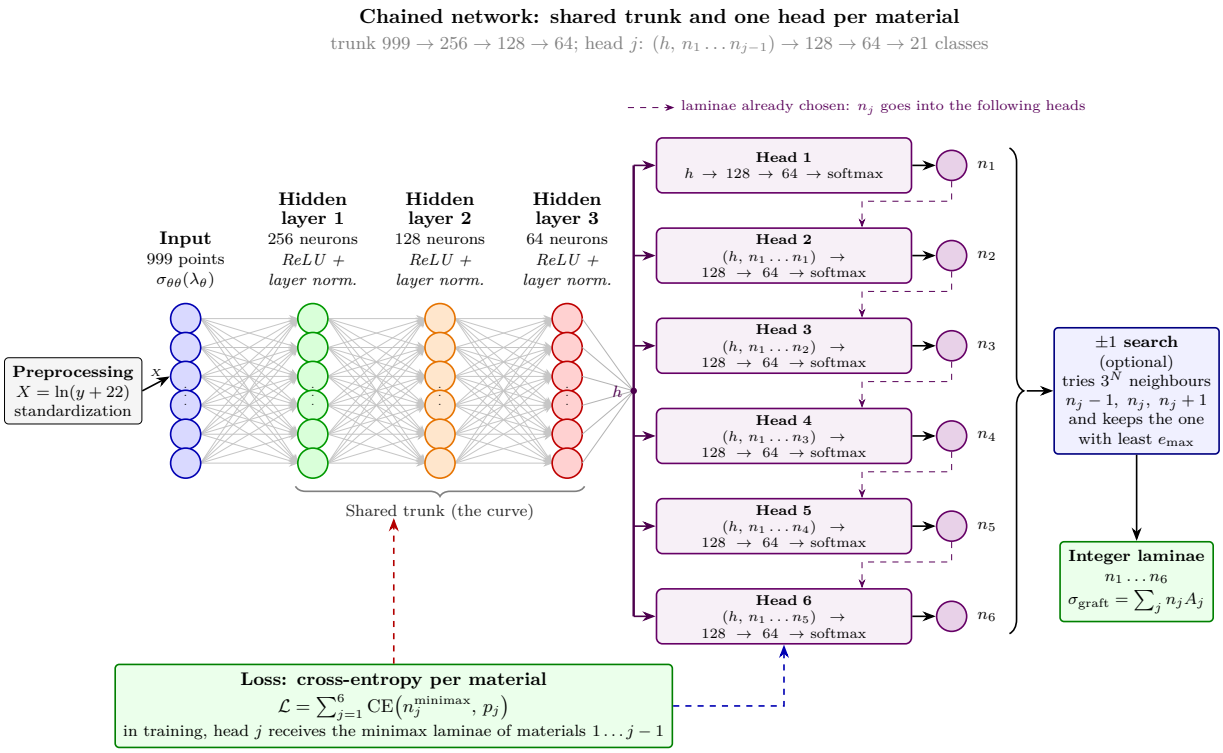


Figure 3: Chained network for 6 materials. A shared trunk ($999 \to 256 \to 128 \to 64$, ReLU and layer normalization) summarizes the curve in the vector h; head j receives h and the laminae $n_1 \dots n_{j-1}$ already chosen by the previous heads (dashed violet arrows), and chooses n_j between 0 and 20 laminae. The loss is the cross-entropy of each material against the minimax integer. The optional ± 1 search reviews the 3^N neighbouring combinations of the answer.

Foundation of the chained network. It rests on four results.

(a) *Universal approximation.* A layered network with enough neurons approximates any continuous function on a bounded set [12, 24, 33]. What each head learns are the probabilities $p_{ij}(n)$ of using n laminae of material j for curve i, which vary continuously with the curve; the step to an integer is taken only at the end, by choosing the most probable class.

(b) *Chain rule of probability.* The probability of a complete design always factorizes, with no assumption, as

$$p(n_1, \dots, n_N \mid \mathbf{y}) = \prod_{j=1}^{N} p(n_j \mid \mathbf{y}, n_1, \dots, n_{j-1}), \tag{4}$$

and each factor is what one head computes. This is the idea of autoregressive models [31] and of classifier chains [41]. Choosing each material separately would assume $\prod_j p(n_j \mid \mathbf{y})$, that is, independent materials; since the materials compensate each other, that assumption is false.

(c) *Maximum likelihood.* Each head turns its scores $z_{ij}(n)$ into probabilities with a softmax, $p_{ij}(n) = e^{z_{ij}(n)} / \sum_{m=0}^{K-1} e^{z_{ij}(m)}$, with $K = 21$ classes. The loss is the cross-entropy summed over materials, plus an L2 regularization of the trunk weights:

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{i=1}^{B} \sum_{j=1}^{N} \Big[-\ln p_{ij}\big(n^*_{ij}\big) \Big] + \lambda \sum_{W \in \text{trunk}} \|W\|^2, \qquad \lambda = 10^{-5}, \tag{5}$$

where n^*_{ij} are the minimax laminae and $B = 256$ curves per batch. By (4), the sum over j is exactly $-\ln p(n^*_{i1}, \dots, n^*_{iN} \mid \mathbf{y}_i)$: minimizing $\mathcal{L}$ is maximizing the probability the network

assigns to the minimax design. To train each factor of (4), head j receives the minimax laminae of the previous materials (*teacher forcing* [53]); in use it receives those chosen by the previous heads, $\hat{n}_{ij} = \arg\max_n p_{ij}(n)$.

(d) *Shared representation h.* The N heads solve tasks that depend on the same curve; a shared trunk forces h to keep what serves all of them and improves generalization over separate networks [10, 7]. The curves depend on few parameters: the Masson law uses about 8 per subject, and the linear fit reproduces any curve of the working space with N coefficients and an error below 1 %. The 64 numbers of h are enough with margin. The size 64, like 256 and 128, is a design choice and does not follow from a theorem.

The theory guarantees that such a network exists, not that training finds it: that is checked with the data. Choosing the most probable class in order does not ensure the most probable combination as a whole either, and in use a head may receive an error it never saw in training (*exposure bias* [6]).

± 1 search. The answer of the network can be reviewed: the 3^N combinations that differ by at most one lamina per material (729 with 6 materials, staying within 0 to 20) are tried, and the one with the least $e_{\max}$ against the curve is kept. It only evaluates Az for each combination: there is no optimization.

3.6 Domain of validity and training

Domain of validity. The network is trained within the y_1–y_2 band of the Masson population (Section 2.2), and only there can the quality of its design be ensured: a curve that leaves the band is outside what the network learned. That band is the *domain of validity* of the method, set with the Masson subjects before looking at any external curve. Before using the network, two conditions are checked: that the curve lies within y_1–y_2 over the whole λ_θ domain, and that it is J-shaped (*class check*: it must admit a non-negative approximation with 40 truncated lines with NRMSE $\leq 2\,\%$, by non-negative least squares [32]). A curve that fails the second cannot be reproduced with layers; one that fails the first is outside the reach of the network, and for it only the minimax applies.

Data. 50 000 curves are generated in seven families: three of the Masson law (parameters of 2 to 4 training subjects mixed and slightly perturbed; mixed with weights that allow leaving the range of the subjects somewhat; or drawn between the minimum and maximum of each parameter), another four-parameter exponential law fitted to the subjects, the 31 training subjects, and two families of sums of laminae of the basis itself (of one material or several). Only those lying entirely within y_1–y_2 are kept: 34 503 with 6 materials and 34 624 with 5. All Masson and subject curves are kept, 92 % of the exponential ones, half of the multi-material sums and only 1 % of the single-material ones, which almost always fall outside the band. The input carries 1 % Gaussian noise; the label of each curve is its minimax laminae. Partition stratified by family: 24 151 for training, 3451 for validation and 6901 for testing (6 materials; 24 236, 3463 and 6925 with 5).

Training. Adam [30] with gradient clipping, batches of 256, learning-rate reduction on plateau and early stopping [39]; TensorFlow/Keras [2]. The 6-material network stopped at epoch 479 (weights of epoch 419) and the 5-material one reached the maximum of 800 epochs with the validation loss already stable, in 9 and 15 minutes of CPU. The learning curves and losses are in Appendix C.

3.7 Results on the Masson population

Results are reported separately for the 31 training subjects, which the network saw (with noise) during training, and for the 10 test subjects, which took part in no choice of the method. 2000 random curves of the working space are added, with a 4 % target (Table 2).

Table 2: Summary of results with integer laminae: median / worst $e_{\max}$ [%] · curves meeting the target (3 %; 4 % for the 2000 working-space curves). All errors against the original curve. External validation: the curves from other authors inside y_1–y_2 (Section 3.8).

	Set	Minimax (reference)	Chained network	Chained network + ±1
6 materials	Masson, training (31)	1.01 / 2.34 · 31	1.47 / 2.90 · 31	1.13 / 2.34 · 31
	Masson, test (10)	1.04 / 1.61 · 10	1.49 / 2.82 · 10	1.04 / 1.61 · 10
	Working space (2000)	1.06 / 3.28 · 2000	1.38 / 9.34 · 1988	1.13 / 4.25 · 1999
	External validation (63)	1.91 / 5.71 · 57	2.26 / 7.71 · 44	2.00 / 5.71 · 53
5 materials	Masson, training (31)	1.48 / 3.50 · 30	1.77 / 3.50 · 30	1.58 / 3.50 · 30
	Masson, test (10)	1.32 / 2.03 · 10	1.62 / 2.35 · 10	1.32 / 2.03 · 10
	Working space (2000)	1.44 / 3.48 · 2000	1.77 / 8.01 · 1984	1.49 / 5.48 · 1999
	External validation (63)	2.29 / 5.98 · 54	2.86 / 8.69 · 33	2.40 / 5.98 · 50

(i) **Test subjects.** With 6 and with 5 materials, the chained network meets the target for the 10 test subjects, alone (worst 2.82 and 2.35 %) and with the ±1 search, which for all 10 reaches the same worst case as the minimax (1.61 and 2.03 %).

(ii) **Training subjects.** With 6 materials all 31 meet it with the three methods. With 5, subject NT04 stays at 3.50 % also with the minimax: there is no 5-material integer design that brings it to the target, and the limit is having only 5 materials, not the network.

(iii) **Different laminae, almost the same curve.** The network gives exactly the minimax laminae for 6 of the 41 subjects (6 materials) and for 16 (5); with the ±1 search, for 13 and 28. Where it differs, it does so by one or two laminae with similar errors: several integer combinations reproduce almost the same curve. Since the minimax minimizes the worst point and not the mean error, the network has an E_{rms} lower than or equal to that of the minimax for 22 of the 41 subjects (6 materials). The laminae and errors of each subject are in Tables 12 and 13 (Appendix D).

(iv) **Edges of the band.** On the 2000 working-space curves, the network alone leaves 12 outside the target (6 materials) and 16 (5); with the ±1 search, one in each case. The hardest curves for the network are those at the edges of the band, where there are fewer training curves: at y_1 and y_2, with 6 materials, the network alone errs by 5.77 and 3.64 %, and with the ±1 search it reaches 1.10 and 1.00 %, the same or almost the same design as the minimax (Fig. 4, Table 3).

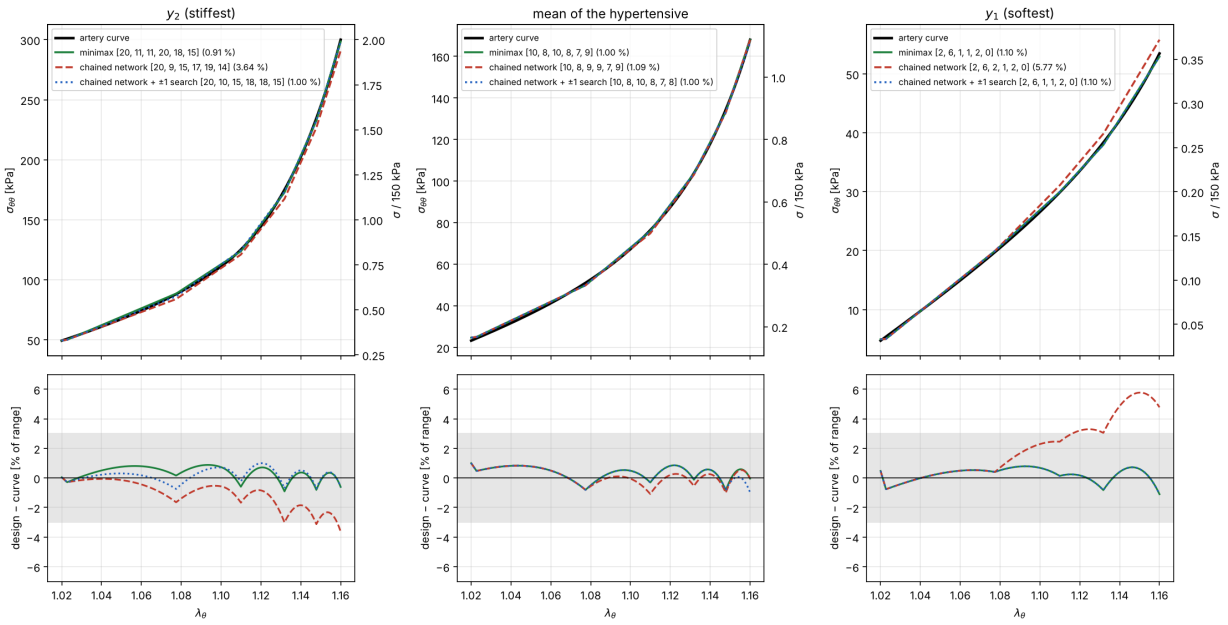


Figure 4: 6 materials: the stiffest curve of the band (y_2), the mean of the hypertensive subjects and the softest (y_1). Top: stress (left axis in kPa, right axis relative to 150 kPa), with the laminae of each material and the $e_{\max}$ of each design; bottom: design − curve in % of the range (gray band: ±3 % target).

Table 3: Designs of Fig. 4 (6 materials): laminae of each material c_j (real-valued in the first row of each curve, integer in the others) and errors against the original curve, E_{rms} and $e_{\max}$ [% of the range].

Curve	Method	c_1	c_2	c_3	c_4	c_5	c_6	E_{rms}	$e_{\max}$
y_2	Least squares (real-valued)	19.86	10.33	12.85	20.00	17.78	14.58	0.32	0.94
	Minimax (MILP, reference)	20	11	11	20	18	15	0.54	0.91
	Chained network	20	9	15	17	19	14	1.39	3.64
	Chained network + ±1 search	20	10	15	18	18	15	0.42	1.00
Mean HT	Least squares (real-valued)	9.43	8.86	8.54	8.17	7.96	7.59	0.30	0.71
	Minimax (MILP, reference)	10	8	10	8	7	9	0.54	1.00
	Chained network	10	8	9	9	7	9	0.51	1.09
	Chained network + ±1 search	10	8	10	8	7	8	0.54	1.00
y_1	Least squares (real-valued)	2.08	5.72	1.41	1.32	1.18	1.07	0.16	0.90
	Minimax (MILP, reference)	2	6	1	1	2	0	0.49	1.10
	Chained network	2	6	2	1	2	0	2.76	5.77
	Chained network + ±1 search	2	6	1	1	2	0	0.49	1.10

Time. The minimax takes about 50 to 65 ms per curve and does not speed up in batch, because it solves one problem per curve. The network, in batch, takes 0.3 ms per curve (0.2 with 5 materials), and with the ±1 search 2.3 ms (0.7 with 5): 20 to 200 times less. For a single curve the time is set by the call to the network library (about 0.3 s with TensorFlow), not by the computation. The advantage of the network is not only time: it gives the design without solving any optimization, with the same operation for any curve in the domain of validity.

3.8 External validation: measurements by other authors

To verify that the method does not depend on the artery with which it was built, it was applied to $\sigma_{\theta\theta}(\lambda_\theta)$ curves from three independent studies, with other constitutive laws, other arteries, isolated layers and in vivo measurements. **Nothing was changed**: the same knots and slopes (Table 1),

the same network, the same domain of validity and the same minimax. None of these laws is part of the training of the network.

Curves and constitutive laws. Each curve is recomputed with the constitutive law and the parameters published by its authors. The law is used as is because it is the one those authors fitted to their measurements: it sets the shape of the curve and, with it, what the graft has to reproduce.

- **Sommer and Holzapfel** [46]: common (CCA) and internal (ICA) carotid arteries from 11 donors (mean age 77.6 years), tested by extension–inflation as intact-wall, adventitia and media–intima tubes [45]. The authors fitted to each specimen the Holzapfel, Gasser and Ogden law [23] with fiber dispersion,

$$\Psi = \tfrac{\mu}{2}(I_1 - 3) + \tfrac{k_1}{k_2}\big(e^{Q} - 1\big),$$
$$Q = k_2\big[(1-\rho)(I_1-3)^2 + \rho\,(I_4 - 1)^2\big], \qquad I_4 = \lambda_\theta^2 \cos^2\varphi + \lambda_z^2 \sin^2\varphi,$$

  where μ is the stiffness of the matrix (elastin), k_1 and k_2 the stiffness and stiffening of collagen, φ the fiber angle and ρ the degree of fiber alignment; the fibers carry no load in compression ($I_4 < 1$). The five parameters of each specimen are in their Tables 1 and 2. We use $\sigma_{\theta\theta} - \sigma_{rr}$ with $\lambda_z = 1.10$, the in vivo axial stretch adopted by the authors [45].

- **Liu et al.** [34]: 12 human carotid arteries with plaque, characterized in vivo with cine and multi-contrast magnetic resonance imaging. The authors identified for each patient a modified Mooney–Rivlin law in uniaxial tension,

$$\sigma = 2\lambda\big(\lambda - \lambda^{-2}\big)\Big(c_1 + D_1 D_2\, e^{D_2(\lambda^2 + 2/\lambda - 3)}\Big), \qquad D_2 = 2,$$

  with c_1 and D_1 from their Table 1. They also report the circumferential shrinkage S between the unloaded state and diastole and the diastolic and systolic perimeters, which give the measured window.

- **Kamenskiy et al.** [29]: 16 patients, common, external (ECA) and internal carotid arteries measured in vivo by ultrasound over the cardiac cycle (48 curves). For each artery they report the stress and the tangent modulus at diastole and at systole and the diameters. The simplest law that reproduces these data with a J shape is the exponential, $\sigma(\varepsilon) = \sigma_d + \alpha\big(e^{\beta\varepsilon} - 1\big)$ on $\varepsilon \in [0, \varepsilon_s]$: with $\sigma(\varepsilon_s) = \sigma_s$ exact and β reproducing the two slopes. In all 48 arteries the systolic slope exceeds the diastolic one: all are J-shaped.

The rabbit carotid artery of Holzapfel et al. [23] was excluded: as isolated layers with the rabbit geometry, at these pressures it gives stresses of 900 to 3300 kPa, far from any human carotid artery.

A single scaling: the pressure window. Each curve lives in its own range of λ and is brought to [1.02, 1.16] by an affine change of λ: the stretch at the low pressure goes to 1.02 and the one at the high pressure to 1.16. **The stress is not touched**: each curve keeps its own stresses, with no value imposed at the ends. The change of λ is equivalent to what the real graft allows: a prestretch (shift of the knots, the δ of Section 4) and the choice of the reference radius. The window comes from the pressure:

- **M (in vivo, measured):** Liu and Kamenskiy, between the diastole and the systole of each patient.
- **P (ex vivo, computed):** Sommer and Holzapfel, between 60 and 200 mmHg, the same pressure window as at the tube level (Section 4). The stretch at each pressure follows from the thin-wall equilibrium used by Sommer et al. [45] (their Eqs. 1 and 2),

$$\sigma_{\theta\theta} = p\Big(\frac{r}{h} - 1\Big), \qquad r = R\,\lambda_\theta, \qquad R^2 - (R-H)^2 = \big[r^2 - (r-h)^2\big]\lambda_z,$$

with the mean unloaded outer radius R and thickness H of each artery and layer (their Table 2): intact wall 4.15 and 1.17 mm (CCA) and 2.67 and 0.86 mm (ICA); adventitia 4.00 and 0.47 and 2.51 and 0.33 mm; media–intima 3.61 and 0.70 and 2.15 and 0.53 mm. Three adventitias (CCA II l and VIII r, ICA VIII r) already carry more than 60 mmHg unloaded and have no window: they are excluded.

At 60 mmHg the ex vivo curves start between 21 and 92 kPa and at 200 mmHg they reach between 76 and 325 kPa; the in vivo curves span 13 to 389 kPa. In total, 98 curves: 38 from Sommer and Holzapfel, 12 from Liu and 48 from Kamenskiy (Fig. 5).

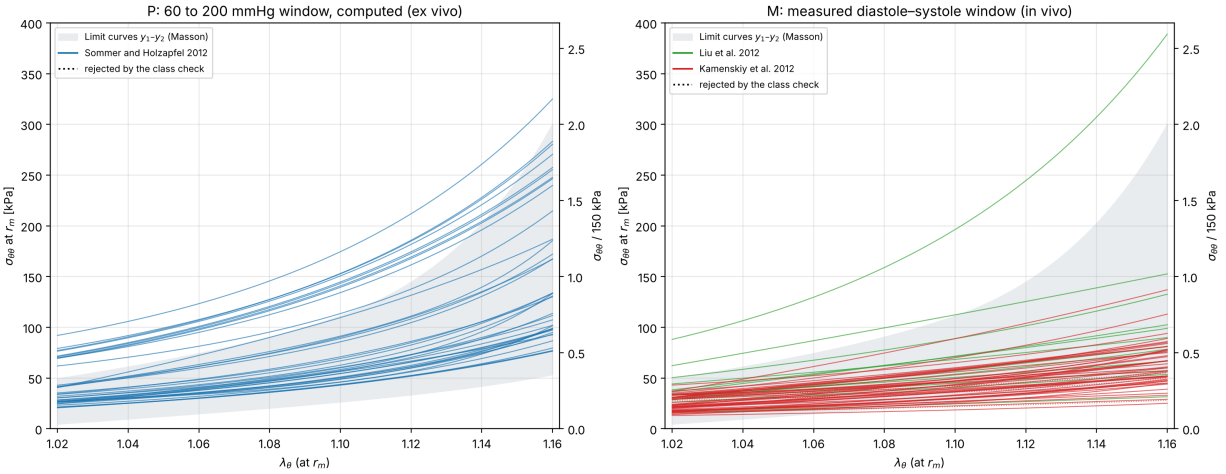


Figure 5: The 98 external curves in the design domain, with the scales of Fig. 1: left axis in kPa, right axis relative to 150 kPa (here up to 400 kPa, because one curve of Liu et al. reaches 389 kPa). Both panels use the same scaling: each curve between its low and its high pressure, with its own stresses. Left: 60 to 200 mmHg window computed with the geometry of each artery and layer (Sommer and Holzapfel); right: diastole–systole window measured in vivo (Liu, Kamenskiy). Gray band: limit curves y_1–y_2 of the Masson population, the domain of validity of the method; only the curves lying entirely inside it make up the external validation. Dotted: rejected by the class check.

Curves inside the domain of validity. Of the 98 curves, the class check rejected 3: the three shortest in vivo windows of Kamenskiy (ECA of patients 11 and 15 and ICA of patient 15, with $\varepsilon_s \leq 3\,\%$). Of the remaining 95, 63 lie entirely within y_1–y_2 and make up the external validation: the 28 intact walls and media–intima of Sommer and Holzapfel, 8 curves of Liu and 27 of Kamenskiy. The 32 that leave the band are not used, because the network cannot ensure the target on them: the 10 isolated adventitias, which exceed y_2 at 200 mmHg; 4 of Liu (P11 reaches 389 kPa); and 18 of Kamenskiy.

Results. Table 4 summarizes the 63 curves by study; Fig. 6 shows the $e_{\max}$ of each curve, and Tables 14 and 15 (Appendix D) the laminae and errors of each one.

Table 4: External validation: the 63 curves inside y_1–y_2. Curves meeting the 3 % target with integer laminae and worst $e_{\max}$ [%] of each method.

		6 materials			5 materials		
Study	Curves	Minimax	Network	Network $+\pm1$	Minimax	Network	Network $+\pm1$
Sommer and Holzapfel [46] (wall and media–intima)	28	28 · 2.52	24 · 4.22	28 · 2.64	28 · 2.76	21 · 4.42	27 · 3.24
Liu et al. [34]	8	5 · 3.74	5 · 6.24	5 · 3.74	5 · 3.91	3 · 8.69	5 · 3.91
Kamenskiy et al. [29]	27	24 · 5.71	15 · 7.71	20 · 5.71	21 · 5.98	9 · 7.29	18 · 5.98
Total	63	57	44	53	54	33	50

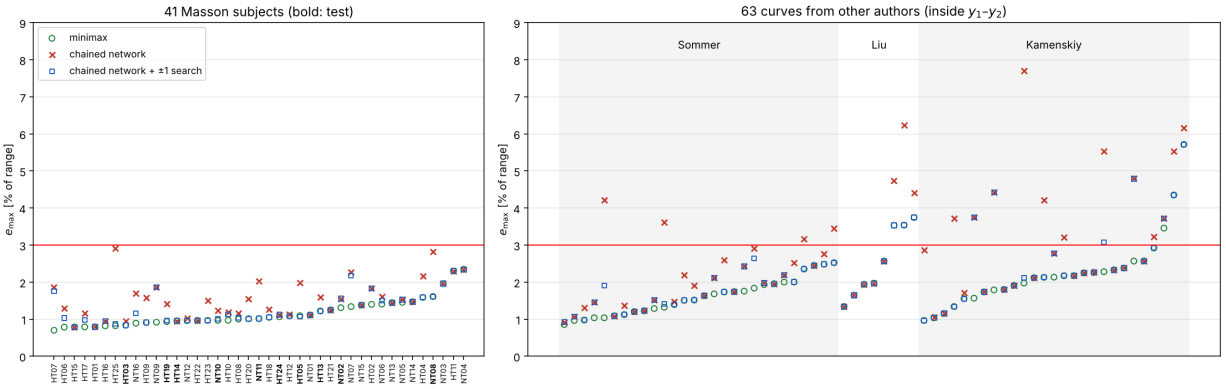


Figure 6: Worst error of each curve with integer laminae, 6 materials: minimax (circles), chained network (crosses) and chained network with ±1 search (squares). Left: the 41 Masson subjects, sorted by the minimax error (test subjects in bold). Right: the 63 external curves inside y_1–y_2, by study. Red line: 3 % target.

(i) **The basis suffices.** With least squares, all 63 curves meet the target: worst error of 1.28 % with 6 materials and 1.84 % with 5.

(ii) **The ceiling of integer laminae.** The minimax meets the target for 57 of the 63 (6 materials) and 54 (5). Those that fail even with the minimax are in vivo curves with a small stress span (26 to 36 kPa with 6 materials, against a median of 56 kPa over the 63), almost straight, in which the first material carries almost the whole design: one lamina more or less moves the curve by more than the target allows, which for them is 1 kPa or less. The worst minimax error on those curves is 1.0 to 1.7 kPa. With real-valued amounts all of them meet it: the limit is set by the division into 20 laminae per material, not by the basis or the network.

(iii) **The network.** Alone, it meets the target for 44 of 63 (6 materials) and 33 (5); with the ±1 search, for 53 and 50. For the 28 curves of Sommer and Holzapfel, with 6 materials, the network with the ±1 search meets it for all, like the minimax; for Liu, for the same 5 as the minimax. Where it falls below the minimax —4 Kamenskiy curves with 6 materials— the network erred by more than one lamina, and a ±1 review is not enough. It gives exactly the minimax design for 14 curves (24 with the search).

Overall result. Adding the 41 subjects and the 63 external curves (6 materials), the minimax meets the target for 98 of 104: that is the ceiling, because no integer design meets it for the other 6. The chained network with the ±1 search meets it for 94, 96 % of what is achievable with integer laminae, without solving any optimization. By set: 31 of 31 training subjects, 10 of 10 test subjects and 53 of 63 external curves.

Conclusion of the first level

With fiber-like materials and a basis chosen once, the stress–stretch curve of the 41 carotid arteries is reproduced with integer laminae within the target (6 materials, at most 20 laminae per material). The minimax gives the best integer design that exists and is the reference. A single neural network, the chained network, trained inside the y_1–y_2 band —its domain of validity—, directly gives the integer laminae in one evaluation: with a ±1 lamina search it meets the target for the 10 test subjects and for 53 of 63 curves from three other studies, with other constitutive laws, and reaches 94 of the 98 cases that admit an integer design within the target. Outside the band the network is not used.

4 Application: the pressure–diameter curve and the thick-walled tube

4.1 From stress to pressure

A graft is implanted as a tube, and what must match the artery is its diameter at each pressure. Going from stress to pressure requires the geometry and an assumption about the wall. The stress-free configuration of the artery is an open annular sector with half-angle Θ_0 and radius R_m at the media–adventitia boundary, which represents the residual stresses (Fig. 7). A point at radius R in the sector moves to radius r in the closed artery; its stretch and volume conservation give [35]

$$\lambda_\theta = \frac{\pi r}{\Theta_0 R}, \qquad \Theta_0\,(R_m^2 - R^2) = \pi\lambda_z\,(r_m^2 - r^2). \tag{6}$$

Radial equilibrium integrated over the thickness gives the pressure borne by the intima–media wall [35, 27]:

$$P = P_i - P_a = \int_{r_i}^{r_m} \frac{\sigma_{\theta\theta} - \sigma_{rr}}{r}\, dr, \tag{7}$$

where P_i is the blood pressure and P_a the pressure exerted by the adventitia and the tissue surrounding the artery. Since P_a depends on the site, the design is formulated in terms of P, the pressure borne by the wall, which the graft must bear with the same deformation. The intima–media thickness h is taken from the age- and sex-specific reference median measured with echotracking [15] (0.43 to 0.68 mm), because Masson et al. do not report it for each subject, and $r_i = r_m - h$.

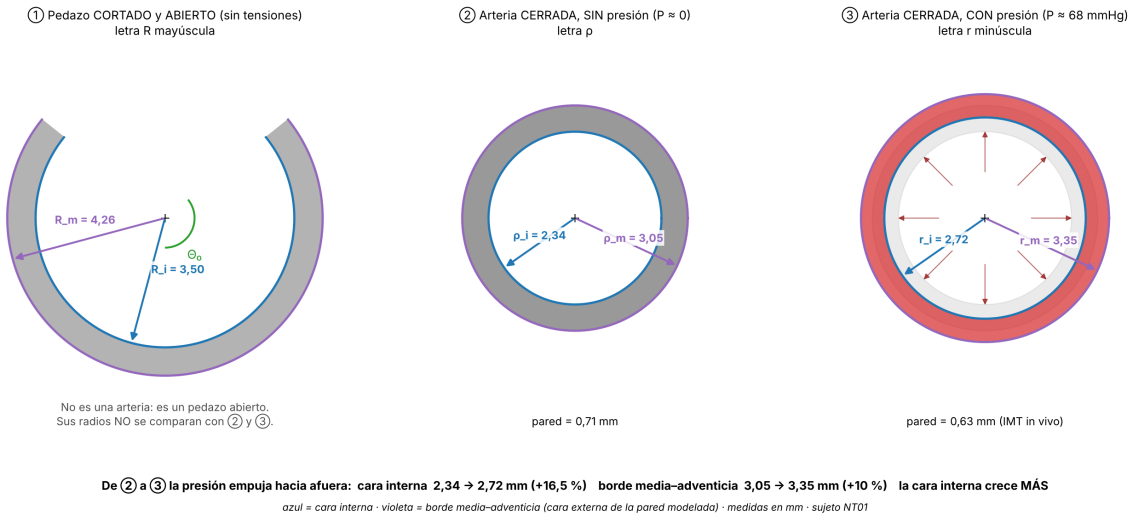


Figure 7: Subject NT01 in its three states: (1) stress-free open sector, (2) closed unpressurized artery, (3) pressurized artery. From (2) to (3) the inner surface grows by 16.5 % and the media–adventitia boundary by 10 %.

Pressure window. What defines whether a state is physiological is the pressure, not the stretch: at $\lambda_\theta = 1.16$ the stiffest subject bears about 400 mmHg and a soft young subject about 70. Therefore, at this level, all subjects are evaluated in the same window, P from 60 to 200 mmHg (8.0 to 26.7 kPa) at 141 values, which covers with a margin the measured mean pressures (64/116 mmHg in NT and 73/127 mmHg in HT [35]). The target curve is the stretch of the media–adventitia boundary, $\lambda(P_k)$, proportional to the diameter (Fig. 8): stiff subjects reach 200 mmHg with λ between 1.12 and 1.15, soft ones with λ between 1.20 and 1.23.

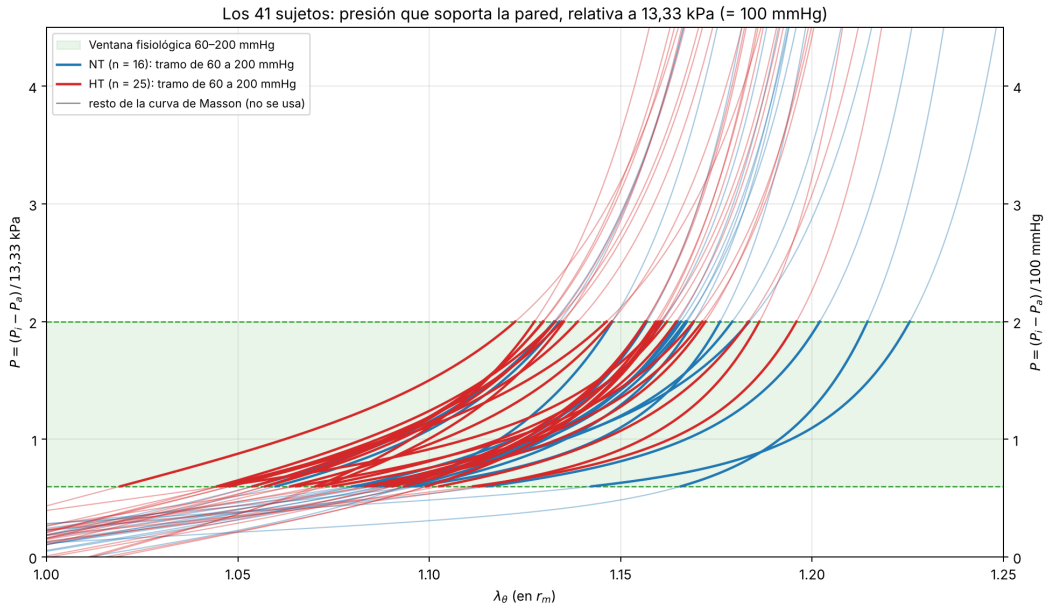


Figure 8: The 41 subjects: pressure borne by the wall as a function of the stretch of the media–adventitia boundary, in relative form: $P = (P_i - P_a)/13.33\,\text{kPa}$ on the left and $P = (P_i - P_a)/100\,\text{mmHg}$ on the right. Since $13.33\,\text{kPa} = 100\,\text{mmHg}$, both scales coincide; the 60 to 200 mmHg window spans 0.6 to 2. Thick lines: the 60 to 200 mmHg segment, which is the target curve.

4.2 Thin wall and thick wall

If the wall is thin compared with the radius, all its fibers stretch equally and the equilibrium of half a tube gives Laplace's law, $P = \sum_j \sigma_j h_j / r$ [17]: pressure is linear in the layer thicknesses and the design inherits the whole method of the first level. Urquiza [51] (Eq. 3.13 of his thesis) writes the same thin-wall equilibrium with the *instantaneous* radius and thickness, $(P_i - P_a)\,R = \sigma\,h$. Taking R as the deformed mean radius, $\bar{r} = r_i + h/2$, and letting $u = 1 + h/r_i$,

$$P = \frac{\sigma\,h}{r_i + h/2} = \sigma\,\frac{2\,(u-1)}{u+1}. \tag{8}$$

For a homogeneous wall with uniform stress, (7) gives exactly $P = \sigma \ln u$, and $2(u-1)/(u+1)$ is the Padé [1/1] approximant of $\ln u$: for $h/r_i \le 0.2$ the error is below 0.3 %, whereas Laplace's law with the inner radius, $P = \sigma\,(u-1)$, overestimates the pressure by up to 10 %. We call *Urquiza's law* Eq. (8) applied to the multilayer graft with a single stretch, that of the mean radius: $P = \sum_j \sigma_j(\bar{\lambda})\,h_j/\bar{r}$, with $\bar{\lambda} = \bar{r}/\bar{R}$. It is still linear in the thicknesses. The carotid artery, however, has $h/r \approx 0.2$. With an incompressible wall and no axial extension, the area between the inner surface and a fiber initially at radius R is conserved,

$$r^2 - r_i^2 = R^2 - R_i^2, \qquad \lambda(r) = \frac{r}{R}, \tag{9}$$

and inner fibers stretch more than outer ones (Fig. 9). The pressure is the exact sum of the contributions of each thin layer, each with its own stretch:

$$P = \sum_{j=1}^{N} \int_{\text{layer } j} \frac{\sigma_j(\lambda(r))}{r}\,dr. \tag{10}$$

If $h \ll r$, (10) reduces to Laplace's law. With truncated lines, the difference in stretch decides whether a layer bears load or not: a material with knot 1.08 bears load at the inner surface of an inflated tube ($\lambda = 1.10$) and not at the outer surface ($\lambda = 1.07$). Moreover, the layers interact: if an inner layer grows, the outer ones are pushed to larger radii, stretch less and contribute less. In a thick wall, pressure is no longer linear in the thicknesses.

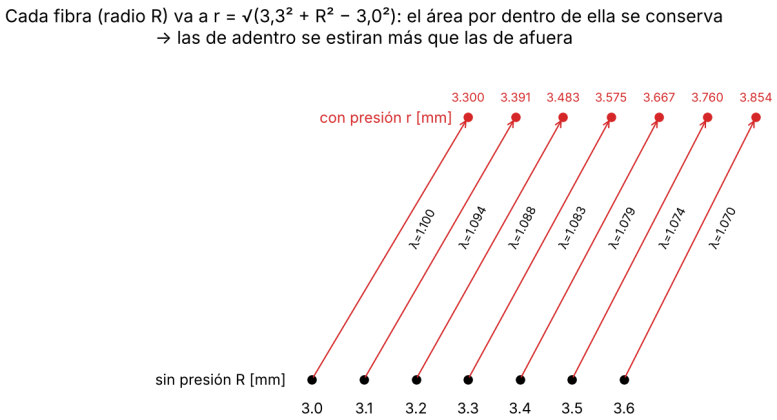


Figure 9: Tube from 3.0 to 3.6 mm inflated to an inner radius of 3.3 mm: each fiber conserves the area enclosed inside it; the inner surface stretches 10 % and the outer surface 7 %.

The thin wall fails when manufactured. Each subject was designed with three assumptions —Laplace (thin wall, $P = \sigma h/r$ with the undeformed radius and thickness), Urquiza's law (8), with the deformed mean radius, and thick wall (Eq. 10, each layer with its own stretch)— and each design was evaluated in the real, thick-walled tube (Table 5, Fig. 10). In its own model, Laplace appears perfect ($e_{\max} \leq 1.7\,\%$); the error appears upon manufacturing and grows with wall thickness. Correcting the radius does not help, because the problem is not the radius but the fact that each layer stretches differently and engages at a different moment. With the pressure window and the adopted basis, the linear thin-wall design does not meet the target in any subject either (median 27.3 % with 6 materials and 29.9 % with 5). **The tube-level design must be carried out with the thick-walled model.**

Table 5: Comparison of assumptions on pressure curves for λ from 1.02 to 1.16, with a set of knots fitted to those pressure curves (different from that of Table 1). Each row is the assumption used to *design*; all designs are *evaluated* in the thick-walled tube, which is what is manufactured. In each cell, for the 41 subjects: how many meet $e_{\max} \leq 3\,\%$, the median $e_{\max}$ and the worst $e_{\max}$, with 5 or 6 materials.

Design assumption	5 materials	6 materials
Laplace	0/41 · 13.6 % · 22.7 %	0/41 · 14.8 % · 32.5 %
Urquiza (deformed mean radius)	0/41 · 13.7 % · 24.5 %	0/41 · 15.4 % · 34.4 %
Thick wall	41/41 · 0.96 % · 2.97 %	41/41 · 0.72 % · 2.10 %

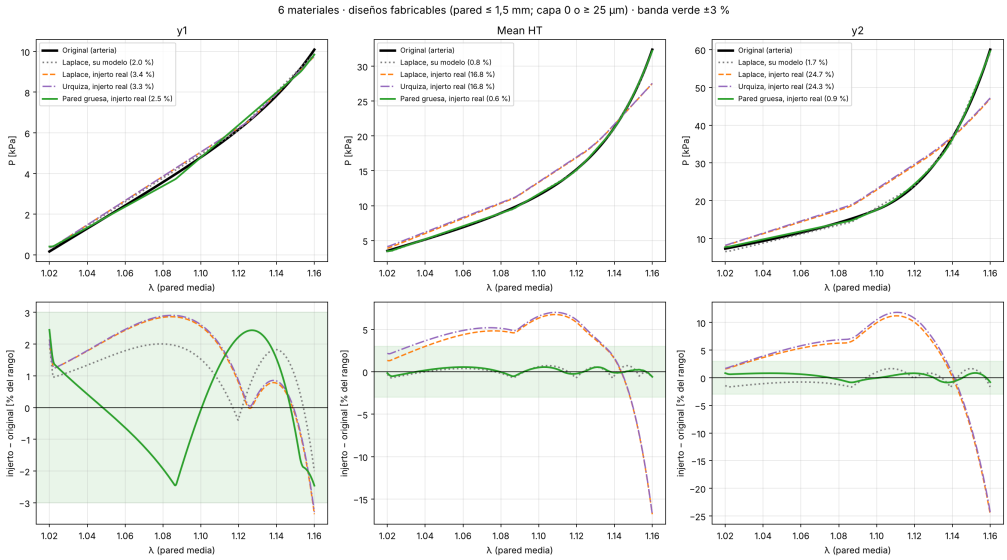


Figure 10: Conditions of Table 5, 6 materials, from left to right: softest curve (y_1), hypertensive mean and stiffest curve (y_2). Top of each pair: curves; bottom: graft − artery in % of the range (band ±3 %). The Laplace design evaluated in its own model (gray, dotted) appears perfect; once manufactured (orange, dashed) it departs by up to 25 %.

Table 6: Designs of Fig. 10 (pressure curves for λ from 1.02 to 1.16, 6 materials; knots 0.8271, 1.0219, 1.0867, 1.1197, 1.1391, 1.1531 and moduli 0.18, 1.33, 4.86, 2.72, 12.94, 6.35 MPa): thickness of the layer of each material h_j [µm] (layer 1 next to the blood; 0: not used), total wall [µm] and errors against the pressure curve of the artery, E_{rms} and $e_{\max}$ [% of the range]. "Laplace, own model" and "Laplace, real graft" are the same design evaluated with two models.

Curve	Design and evaluation	h_1	h_2	h_3	h_4	h_5	h_6	wall	E_{rms}	$e_{\max}$
y_1	Laplace, own model	25	153	0	56	0	0	233	1.47	2.01
	Laplace, real graft	25	153	0	56	0	0	233	1.98	3.36
	Urquiza, real graft	25	153	0	56	0	0	233	2.01	3.29
	Thick wall, real graft	25	130	25	0	0	29	210	1.47	2.46
HT mean	Laplace, own model	235	203	59	205	67	136	905	0.41	0.76
	Laplace, real graft	235	203	59	205	67	136	905	5.14	16.79
	Urquiza, real graft	250	196	62	204	66	132	909	5.40	16.79
	Thick wall, real graft	129	158	48	218	0	467	1020	0.34	0.60
y_2	Laplace, own model	449	229	139	461	193	30	1500	1.08	1.73
	Laplace, real graft	449	229	139	461	193	30	1500	7.95	24.69
	Urquiza, real graft	453	241	141	454	187	25	1500	8.24	24.30
	Thick wall, real graft	203	122	67	626	206	277	1500	0.61	0.87

4.3 The thick-walled graft

The graft is a closed, incompressible tube in plane strain, with one layer per material stacked from inside to outside in order of increasing knot (the reverse order doubles or triples the error), centered at $R_{\mathrm{ref}} = 3.421$ mm (median of $\Theta_0 R_m/\pi$ over the 41 subjects). The materials are those of the first level: the same knots and lines $\sigma_j = E_j \max(0, \lambda - \kappa_j)$, with moduli E_j calibrated once so that the training subject that needs the most of each material uses 0.2 mm of it (6 materials: from 452 to 4966 kPa). The design is

$$\theta = (h_1, \ldots, h_N,\ \delta), \qquad 0 \le h_j \le 3 \text{ mm}, \quad |\delta| \le 0.25. \tag{11}$$

Here h_j is the thickness of the layer of material j and δ is the *common prestretch*: a shift of all the knots together, in units of stretch (dimensionless). Layer j starts working when the mid-wall reaches $k_j + \delta$ instead of k_j:

- $\delta < 0$ advances the engagement of all layers (they are mounted tighter: they bear load earlier);
- $\delta = 0$ keeps the knots of the first-level basis;
- $\delta > 0$ delays it (they are mounted looser: they bear load later).

Meaning of $|\delta| \leq 0.25$. $|\delta|$ is the absolute value of δ, so the condition is equivalent to $-0.25 \leq \delta \leq 0.25$: the knots can be shifted by at most 0.25 to the left or to the right. For example, with $\delta = -0.25$ the knot 1.08 moves to 0.83, and with $\delta = +0.25$, to 1.33. It is a bound of the search, not a material property: it was chosen wider than the whole stretch interval spanned by a subject (from 1.02 to 1.23, i.e. 0.21), so that no subject is excluded by the bound. It is not active: in the 82 designs (41 subjects, 5 and 6 materials) δ remained between -0.04 and $+0.09$. Appendix B shows, with an example that can be checked by hand, what δ does in a two-layer tube.

- **Thicknesses** h_j, manufactured with n_j integer laminae of thickness e.
- **Engagement and prestretch.** Each layer engages when the mid-wall reaches $k_j + \delta$; its local engagement stretch κ_j is the stretch of the layer center at that moment, from (9). In manufacturing, κ_j is the crimp of the fibers of that layer. The common prestretch δ shifts all the knots together: each subject occupies its own stretch interval (from 1.02 to 1.23), and without δ it would see only one or two knots, with a typical error of 11 %.

The pressure (10) is integrated in each layer with 8-node Gauss–Legendre quadrature [13] at 651 values of λ between 0.80 and 1.45, and inverted by linear interpolation to obtain $\lambda_{\mathrm{gr}}(P_k; \theta)$. The map $\theta \mapsto \lambda_{\mathrm{gr}}$ is nonlinear: the thicknesses set the integration limits and the engagements, and δ enters inside the $\max(0, \cdot)$. With integer laminae, finding the n_j is a **mixed-integer nonlinear** problem. The metric is (3) on $\lambda(P_k)$.

4.4 Design methods

Table 7: Computational blocks at the tube level. "Linear" refers to the dependence of the model on the unknowns.

Block	Variables	Linear?	Algorithm	Implementation
A1 Linear LS (thin wall)	real	yes, with fixed δ	least squares by SVD; δ by grid and Brent	`numpy.linalg.lstsq` [21]; `minimize_scalar` [9, 52]
A2 Nonlinear LS (thick wall)	real	no	trust region reflective with bounds (TRF)	`scipy.optimize.least_squares` [8, 52]
A3 Integer minimax (thick wall)	integer laminae, real δ	no	floor/ceiling enumeration, δ by bounded Brent, local search	own code; `minimize_scalar` [9]
A4 Neural network	real	— (one evaluation)	multilayer perceptron; amortized optimization	TensorFlow/Keras [2]

A1 · Linear thin-wall least squares. With Laplace's law and fixed δ, pressure is linear in $c_j = E_j h_j / R_{\mathrm{ref}}$: $P_k \approx \sum_j c_j \max(0, y_k - k_j - \delta)$. For each δ on a grid of 101 values the problem is solved by least squares; the pressure residual is converted to λ with the local slope of the curve; the best δ is refined by Brent's method [9] and $h_j = c_j R_{\mathrm{ref}} / E_j$, clipped to $[0, 3]$ mm. It is the first-level method transferred to the tube.

A2 · Nonlinear thick-wall least squares. It minimizes $\sum_k \left((\lambda_{\mathrm{gr}}(P_k; \theta) - y_k)/\Delta\right)^2$ subject to the bounds (11), by the Trust Region Reflective (TRF) method [8]. At each iteration it computes

the Jacobian of the residuals by finite differences, approximates the problem by its Gauss–Newton quadratic model and takes the step that minimizes it within a *trust region*, a neighborhood where the model is reliable; it enlarges the region if the step succeeded and shrinks it otherwise, and reflects inward the steps that cross a bound. Tolerances 10^{-10}, at most 300 evaluations. It finds a *local* minimum: the result depends on the starting point.

A3 · Integer minimax. It is the first-level minimax extended to the nonlinear case, where no linear program can solve it exactly. Starting from a real-valued design $\theta^\star$ and a reference curve $\tilde{y}$: (1) $n_j \in \{\lfloor h_j^\star/e \rfloor, \lceil h_j^\star/e \rceil\}$; the 2^N combinations are evaluated by $e_{\max}$ against $\tilde{y}$; (2) for the 8 best ones, δ is readjusted within ± 0.03 by bounded Brent; (3) local search [1]: neighbors that add or remove one lamina of a material or move one lamina from one material to another, accepting the best one if it lowers the worst error, up to 20 times. It is a heuristic: it does not guarantee the global integer optimum. Unlike the first level, here a real-valued solution is indeed the starting point: this is the usual strategy for nonlinear integer problems, solving the continuous problem and searching for integers in its neighborhood, and it is reasonable because with 0.025 mm laminae a design has about 46 laminae and one lamina shifts the curve only slightly. The distance of this heuristic from the optimum is measured below (*Distance to the integer optimum*).

A4 · Neural network. A multilayer perceptron with the same trunk as the first-level network and linear outputs, $141 \to 256 \to 128 \to 64 \to N{+}1$: it receives the standardized curve $\lambda(P_k)$ and returns θ. Data: 6000 grafts with randomly drawn designs, whose label is exactly their design, and 9000 synthetic arteries of Masson's law plus the 31 training subjects, whose label is the A2 fit; new noise of 0 to 1 % of the range of λ at each epoch. Split 11 220 / 1530 / 2250; 425 epochs (6 materials) and 629 (5). It is used as *amortized optimization* [4]: a model learned on many solved cases proposes the solution of a new one, and a classical optimizer finishes it. Learning curves and metrics are given in Appendix C.

The procedure. The blocks are chained in three steps (Fig. 11): (1) initialization, A1 (path A) or the network A4 (path B); (2) A2, which fits the design to the curve; (3) A3, which chooses the laminae against the *fitted curve* of step 2. This is the first-level conclusion about noise: step 2 smooths, because the curve of a graft cannot have the jumps of the noise, and step 3 does not chase it.

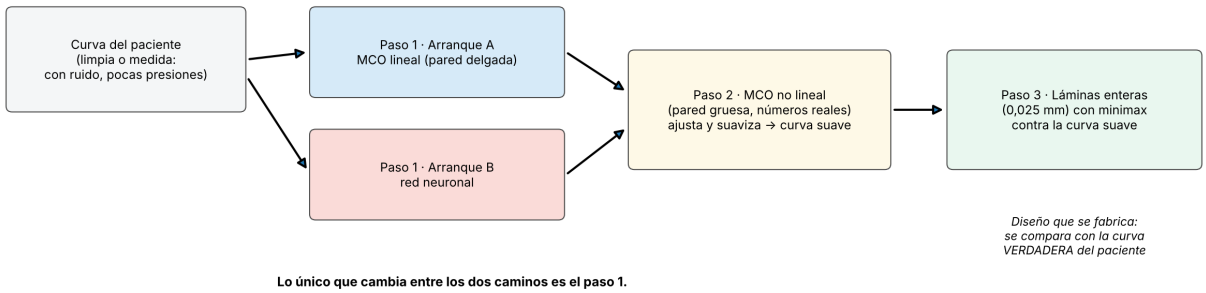


Figure 11: The tube-level design procedure. The two paths differ only in step 1.

Measured curves. In practice the patient's curve is measured, with noise and at few pressures. We simulate: (a) proportional Gaussian noise of 0.3 % in λ (echotracking resolves about 20 µm in 7 mm); (b) measurement at 10 pressures with monotone cubic interpolation (PCHIP) [16] to the 141; (c) both, with 0.2 % noise. Each path receives the measured curve and its integer design is compared with the true curve; one realization per subject and case. Since $\lambda \approx 1.1$ and its range is about 0.07, a noise of 0.3 % of λ corresponds to about 5 % of the range: larger than the target.

4.5 Results

Real-valued design. From the linear initialization, A2 brings 40 of the 41 subjects within the target (6 materials: median 0.84 %, worst 3.43 %; 5 materials: 40/41, worst 3.14 %), with walls from 0.75 to 1.89 mm and δ from -0.04 to $+0.09$. Starting from the network it reaches 41 of 41 with 6 materials (worst 1.79 %), with half the model evaluations (16 versus 35 per subject). On its own, the network falls short: median 3.43 % and worst 89.6 %. In the 10 test subjects the hybrid gives a median of 0.93 % and a worst case of 1.79 %.

Lamina thickness. With laminae of 0.1, 0.05 and 0.025 mm, 38, 40 and 41 of the 41 subjects met the target (6 materials; worst 5.44, 3.69 and 1.64 %; about 10, 23 and 46 laminae). We adopted $e = 0.025$ mm.

The complete procedure. With clean curves both initializations tie (Table 8, Fig. 12). The difference lies in the worst case with measured curves: with 10 pressures, the linear initialization falls into a local minimum in two subjects (NT03 and HT04, 468.7 %), which the network initialization avoids; with 0.3 % noise, the network increases the number of subjects within the target (31 versus 23 with 6 materials; 32 versus 25 with 5). With 10 pressures and noise, none exceeds 8 of 41: that noise is already comparable to the target. The time is 0.2 to 0.5 s per subject with either initialization.

Distance to the integer optimum. To measure how much the A3 heuristic loses, for the 41 subjects (clean curves, linear LS initialization) we enumerated *all* combinations of laminae within ± 2 of those of the real-valued design of step 2 (up to $5^6 = 15\,625$ combinations), each with δ on a grid of 13 values and refined by Brent's method for the best 200, with the same objective as A3 (worst point against the fitted curve). The exhaustive search yields the same design as A3 in 24 of the 41 subjects (6 materials) and in 34 (5 materials); the worst point decreases in 20 subjects (6 materials), on average by 0.15 and at most by 0.44 points, and in 10 subjects with 5 materials (0.14 and 0.43; sometimes with the same laminae and a different δ). Against the true curve, the median goes from 1.13 to 1.03 % (6 materials) and from 1.69 to 1.62 % (5), and neither the worst case (3.26 and 3.21 %) nor the number of subjects meeting the target (40 of 41 in both cases) changes. The heuristic therefore stays within a few tenths of a point of the optimum in that neighborhood, in 0.5 s instead of the 2 min per subject of the exhaustive search (6 materials).

Table 8: Complete procedure with 0.025 mm laminae: worst error against the true curve [%], 41 subjects.

		6 materials			5 materials		
Curve	Initialization	median	worst	meet	median	worst	meet
clean	linear LS	1.13	3.26	40/41	1.69	3.21	40/41
	network	1.14	2.19	41/41	1.68	3.19	40/41
10 pressures	linear LS	1.26	468.7	38/41	1.62	3.13	40/41
	network	1.19	2.27	41/41	1.61	2.70	41/41
noise 0.3 %	linear LS	2.67	11.87	23/41	2.66	7.52	25/41
	network	2.19	6.59	31/41	2.32	9.24	32/41
10 pres. + noise 0.2 %	linear LS	4.36	8.07	8/41	4.62	8.77	6/41
	network	3.78	8.54	8/41	4.75	10.35	6/41

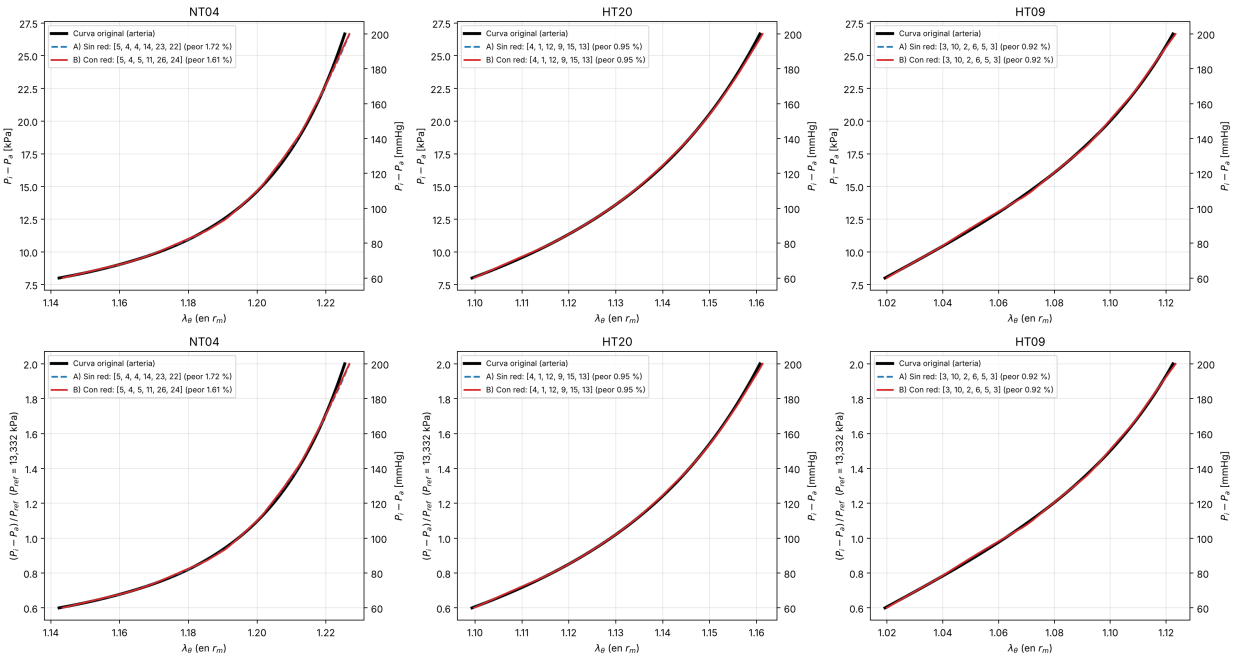


Figure 12: 6 materials, clean curves: the softest subject, an intermediate one and the stiffest one. Black: artery; blue: path A (linear LS initialization); red: path B (network initialization). Ordinate: pressure borne by the wall, in kPa at the top and relative to 13.332 kPa at the bottom; mmHg on the right. In brackets, laminae of each material.

Table 9: Integer designs of Fig. 12 (6 materials, 0.025 mm laminae, clean curves): number of laminae of each material n_j (material 1 next to the blood), total laminae, wall [mm], prestretch δ and errors against the $\lambda(P)$ curve of the artery, E_{rms} and $e_{\max}$ [% of the range]. Path A: linear LS initialization; path B: network initialization.

Subject	Path	n_1	n_2	n_3	n_4	n_5	n_6	laminae	wall	δ	E_{rms}	$e_{\max}$
NT04 (softest)	A	5	4	4	14	23	22	72	1.800	+0.0861	0.59	1.72
	B	5	4	5	11	26	24	75	1.875	+0.0890	0.58	1.61
HT20 (intermediate)	A	4	1	12	9	15	13	54	1.350	+0.0103	0.37	0.95
	B	4	1	12	9	15	13	54	1.350	+0.0103	0.37	0.95
HT09 (stiffest)	A	3	10	2	6	5	3	29	0.725	−0.0394	0.42	0.92
	B	3	10	2	6	5	3	29	0.725	−0.0394	0.42	0.92



Figure 13: 6 materials, measured curves: worst error of each subject against the true curve (logarithmic scale; blue, linear LS initialization; red, network initialization; dotted, target).

Against which curve to choose the laminae. If step 3 chooses the laminae against the curve as measured, it chases the worst point, which with noise is a noise spike (Fig. 14). Against the fitted curve of step 2, with 0.3 % noise and network initialization, 31 subjects meet the target instead of 10 (6 materials) and 32 instead of 5 (5 materials); the cost with clean curves is 0.2 points of median.

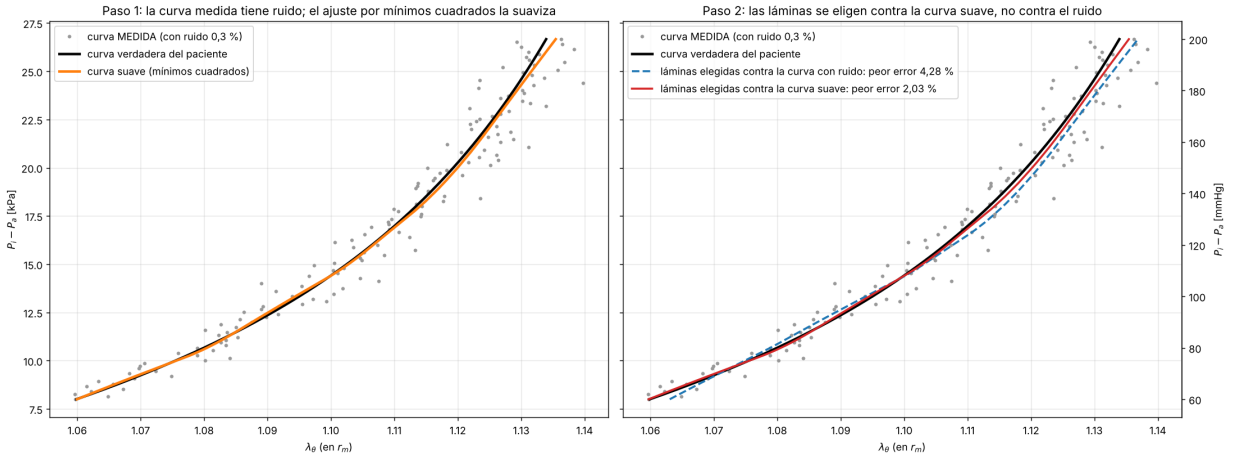


Figure 14: Subject NT11, 6 materials, 0.3 % noise. Left: the curve fitted in step 2 (orange) passes through the middle of the measurement (gray). Right: laminae chosen against the measured curve ($e_{\max} = 4.28\,\%$) and against the fitted one (2.03 %), both evaluated against the true curve.

5 Discussion

Why two levels. The stress–stretch curve is a property of the tissue. Working on it first makes it possible to validate each component of the method —the basis, the fit, the rounding, the network— against exact solutions, without the intervention of geometry or pressure, which depend on each artery and each site. The transition to the tube adds the geometry and the thick-wall assumption, but keeps the materials, the metric, the minimax criterion and the conclusion about noise.

Generality. What is validated at the first level is not specific to the carotid artery: the fiber-like materials, the basis chosen over a broad space of J-shaped curves, the minimax integer design and a network that gives that design for curves of laws it never saw. For another artery, the geometry, thickness and pressure window change, and these only enter at the second level; the band of curves and, if necessary, the knots are recomputed with the same procedure, and the network is retrained inside the new band. The external validation (Section 3.8) confirms this with data: the basis chosen with Masson's carotid arteries reproduces, unchanged, curves of common, external and internal carotid arteries, of intact walls and media–intima and of patients with plaque, from three different constitutive laws, each in its own pressure window. What limits some in vivo curves is the division into 20 laminae per material relative to a small stress excursion, not the shape of the curve or the network.

The role of the neural network. At the first level, the minimax gives the best integer design, but by solving an optimization problem for each curve. The chained network gives the integer design directly, with the same operation for any curve in its domain of validity: alone, it meets the target for the 41 subjects (6 materials); with a ± 1 lamina search, which only evaluates 729 combinations, it reaches 94 of the 98 cases that admit an integer design within the target. It lacks the optimality guarantee of the minimax, which remains the reference to check any design, and it is not used outside the y_1–y_2 band, where it cannot ensure the target. In the thick-wall case the fit is iterative, local and dependent on the starting point: the classical initialization, inherited from the linear case, ignores the stretch gradient across the thickness and, with measured curves, can lead to local minima that yield unusable designs. A network trained on thousands of solved designs leaves the fit close to the correct minimum. This is the idea of amortized optimization [4]: the network does not replace the optimizer, it initializes it. Its advantage is one of robustness, clear with 6 materials and smaller with 5.

Why a thick wall. The difference between the models lies in a single assumption: in a thin wall all fibers stretch equally; in a thick wall each has its own stretch, set by incompressibility. With materials that bear no load until their knot, that difference of a few hundredths decides whether a layer works. The engagement of each layer (its crimp) is as important as its thickness: between a Laplace design and a thick-wall design the thicknesses change by a few micrometers, but the outer layers engage earlier.

Limitations. (1) The target curves are those of the law fitted by Masson et al., not the raw measurements, and the arterial thickness is the age- and sex-specific reference value. (2) The materials are ideal (truncated lines of constant stiffness); the graft has neither axial extension nor residual stresses, and viscoelasticity and fatigue are not considered. (3) The integer search in the thick-wall case is heuristic: it stays within half a point of the optimum in a neighborhood of ± 2 laminae, but the global optimum was not proven; a single noise realization per subject and case was used. (4) The prestretch δ and the per-layer crimp are manufacturing variables that must be validated. (5) The first-level network is valid within the y_1–y_2 band of the Masson population; for curves outside it, the band must be widened and the network retrained, or only the minimax used. (6) There is no experimental validation.

6 Conclusions

1. A graft made of layers of fiber-like materials (truncated lines), with a basis chosen once for the whole population and at most 20 laminae per material, reproduces the stress–stretch curve of 41 human carotid arteries: with 6 materials and integer laminae by minimax, all 41 subjects meet the 3 % target.
2. A single chained neural network, trained inside the y_1–y_2 band —its domain of validity—, directly gives the integer laminae in one evaluation, without optimizing. With a ± 1 lamina search it meets the target for the 31 training subjects and the 10 test subjects, with 6 and with 5 materials, except for one subject that does not meet it with the minimax either.
3. Without any change, the same basis and the same network were applied to 63 curves from three other studies (Sommer and Holzapfel, Liu et al., Kamenskiy et al.) inside the domain of validity, each with its own constitutive law and pressure window: with 6 materials, the network with the ± 1 search meets the target for 53 and the minimax for 57; for the other 6 no integer design meets it, because their stress span is small relative to the step of one lamina. Over the 104 cases, the network reaches 94 of the 98 achievable with integer laminae.
4. At the tube level the design must be carried out with the thick-walled model: designed with Laplace's law, no subject meets the target once manufactured; with the thick-walled model, 40–41 of 41 do. The problem is a nonlinear integer one.
5. The three-step procedure —initialization, nonlinear least squares and integer laminae by minimax against the fitted curve— meets the target in 40–41 of 41 subjects with clean curves, in less than half a second per subject.
6. In the thick-wall case, the network as a starting point avoids the local minima that, with measured curves, produce unusable designs (worst case from 468.7 to 2.3 %) and increases the number of subjects within the target under noise. This is the justified role of the network: to initialize the optimizer, not to replace it.

Next steps. Test the method with raw measured curves of other arteries; test real materials to replace the ideal lines and validate crimp as a manufacturing variable; use each subject's measured thickness and pressure; and manufacture and test a graft.

A Parameters of the 41 subjects

Table 10: Parameters of Masson et al. [35] (their Tables 2 and 4). †: subject held out for testing.

Subject	Age [years]	Sex	R_m [mm]	Θ_0 [°]	λ_z	c [kPa]	c_1 [kPa]	c_2	c_1^{circ} [kPa]	c_2^{circ}	α [°]	T_m [kPa]	λ_0	λ_m	a [kPa]	b
NT01	62	F	4.26	128.70	1.11	29.82	9.44	14.14	16.13	15.11	65.74	39.73	0.960	1.70	0.122	3.52
NT02†	60	M	4.69	128.90	1.08	20.42	8.47	13.10	14.16	15.12	44.46	41.84	0.680	1.92	0.126	3.40
NT03	31	M	4.30	132.20	1.09	27.23	10.34	13.10	7.01	12.82	48.62	43.37	0.820	1.65	0.139	3.57
NT04	31	M	4.67	133.90	1.10	23.59	6.58	11.77	6.38	10.69	32.04	39.81	0.680	1.59	0.118	3.53
NT05	34	F	4.20	130.10	1.10	35.68	13.62	13.18	13.38	13.17	50.72	50.07	0.860	1.62	0.103	3.58
NT06	53	F	3.96	130.10	1.11	33.37	11.38	13.63	13.30	14.26	57.14	46.59	0.950	1.69	0.093	3.44
NT07	21	F	3.83	132.00	1.10	44.28	21.79	13.17	8.27	12.73	51.15	53.58	0.920	1.62	0.111	3.58
NT08†	25	M	4.86	133.20	1.07	23.25	6.34	12.93	8.11	13.17	42.93	31.90	0.820	1.77	0.173	3.57
NT09	42	F	4.51	129.50	1.10	34.90	16.14	13.58	19.64	13.89	57.21	47.01	0.930	1.65	0.084	3.55
NT10†	55	M	4.77	129.10	1.10	38.57	36.31	13.31	56.88	13.61	54.22	46.36	0.890	1.66	0.006	3.75
NT11†	64	M	5.57	129.50	1.10	34.89	40.16	13.45	66.88	13.53	54.54	41.12	0.850	1.79	0.005	3.23
NT12	60	F	4.88	130.10	1.09	33.94	10.78	13.95	12.79	15.42	60.97	45.77	0.930	1.73	0.099	3.37
NT13	61	F	4.92	130.60	1.09	35.15	17.69	13.43	18.40	13.22	55.04	45.59	0.940	1.66	0.085	3.51
NT14	39	F	3.54	131.50	1.09	28.97	8.22	13.13	19.02	13.16	50.22	41.30	0.890	1.67	0.105	3.53
NT15	58	F	3.92	132.40	1.08	26.71	15.79	13.36	16.68	13.11	53.60	35.32	0.950	1.71	0.166	3.60
NT16	60	M	5.72	129.70	1.10	27.79	28.23	13.46	57.77	13.63	55.73	36.50	1.05	1.77	0.141	3.58
HT01	68	M	4.96	130.90	1.10	35.91	12.57	14.08	15.52	14.91	67.18	48.42	0.990	1.66	0.098	3.50
HT02	54	M	4.93	129.80	1.10	38.20	16.40	13.20	16.44	13.20	51.11	52.44	0.860	1.62	0.137	3.58
HT03†	64	F	4.87	129.50	1.10	36.01	14.55	13.96	18.35	14.48	62.66	47.02	1.00	1.74	0.113	3.62
HT04	58	F	4.57	131.20	1.09	22.42	8.57	13.03	6.43	13.36	46.38	38.53	0.810	1.68	0.099	3.51
HT05†	62	F	4.78	130.10	1.10	48.19	35.06	13.23	43.19	13.03	53.24	52.86	0.910	1.63	0.033	3.56
HT06	66	F	5.06	129.10	1.10	39.78	37.75	13.35	60.89	13.45	56.01	46.44	0.910	1.64	0.004	3.54
HT07	54	F	4.30	129.60	1.11	50.45	25.51	13.32	39.43	13.24	53.47	64.91	0.930	1.66	0.078	3.72
HT08	63	F	4.44	129.00	1.10	34.44	30.62	13.29	60.33	13.21	54.67	44.94	0.930	1.65	0.084	3.56
HT09	68	M	5.09	128.20	1.12	51.48	33.77	13.17	44.26	13.49	51.83	66.41	0.840	1.62	0.023	3.55
HT10	55	F	4.30	129.00	1.10	37.04	12.32	14.13	15.04	14.93	65.14	48.19	1.04	1.71	0.114	3.47
HT11	69	M	5.54	131.20	1.08	29.64	9.76	13.23	9.10	15.12	48.16	37.83	0.830	1.74	0.115	3.59
HT12	54	M	5.43	131.80	1.09	32.23	17.81	13.35	14.03	12.15	53.73	44.93	0.920	1.65	0.090	3.58
HT13†	61	M	5.17	128.40	1.11	30.45	15.87	13.08	23.03	13.84	47.86	52.58	0.750	1.61	0.009	3.49
HT14†	60	M	4.64	129.90	1.10	30.43	13.74	13.45	18.01	13.35	54.90	42.43	0.910	1.66	0.088	3.52
HT15	53	M	4.65	130.40	1.10	40.83	28.98	13.36	31.16	12.62	54.44	49.57	0.930	1.65	0.085	3.63
HT16	54	M	4.71	129.60	1.10	45.26	29.27	14.00	33.48	14.82	67.16	67.68	1.02	1.65	0.039	3.52
HT17	53	M	5.04	130.00	1.10	37.25	39.18	13.24	66.29	13.15	54.90	48.26	0.910	1.64	0.099	3.60
HT18	52	M	4.92	129.70	1.10	29.28	10.50	13.21	19.76	13.39	50.96	46.42	0.850	1.63	0.094	3.56
HT19†	57	F	4.27	130.30	1.10	33.49	14.47	13.22	17.35	13.27	53.18	46.18	0.900	1.64	0.097	3.55
HT20	61	F	5.14	129.20	1.10	40.09	15.72	14.09	16.78	14.83	65.17	53.40	1.04	1.70	0.100	3.47
HT21	57	F	4.88	129.90	1.10	35.31	14.57	13.26	15.58	13.30	52.07	48.97	0.870	1.63	0.094	3.57
HT22	44	F	4.17	131.00	1.10	55.25	35.78	13.23	19.51	12.55	52.36	59.58	0.930	1.61	0.097	3.61
HT23	51	M	4.83	130.60	1.08	30.77	12.80	13.60	41.10	14.20	49.79	41.06	0.870	1.90	0.094	3.56
HT24†	65	M	4.57	129.60	1.10	38.91	25.76	13.20	35.01	13.20	51.94	50.28	0.870	1.62	0.043	3.58
HT25	67	F	4.72	129.20	1.10	40.29	31.78	13.75	60.40	14.29	63.50	46.06	1.03	1.72	0.025	3.46

B Worked example of a thick-wall design

This appendix shows, with numbers that can be checked with a calculator, the ideas used in Section 4: the graft reproduces the behavior of the artery with layers of different materials; each fiber of a thick wall stretches differently; pressure is the sum of the contributions of thin slices; the position of each layer changes what it contributes; and the prestretch δ shifts all the knots. To keep the calculations short, the tube is closed, without residual stresses and without axial extension, and the knot of each material is applied directly to the local stretch of each slice (in the complete model the knot is defined at the mid-wall and translated to each layer by incompressibility, Eq. 9).

B.1 The artery and the graft

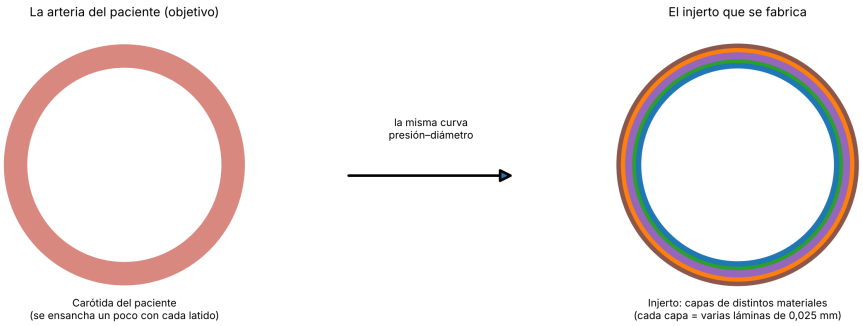


Figure 15: The patient's carotid artery (left) and the graft (right), made of layers of different materials, each layer with an integer number of laminae. The goal is for the graft to have the same pressure–diameter curve as the artery: both should widen equally with pressure.

B.2 Incompressibility: the ring area does not change

Tissue is mostly water: it deforms without changing volume. If the tube does not extend axially, the ring area is the same before and after inflation:

$$\pi\,(R_o^2 - R_i^2) = \pi\,(r_o^2 - r_i^2) \qquad \Longrightarrow \qquad r = \sqrt{r_i^2 + R^2 - R_i^2} \quad \text{for any wall radius } R.$$

Example tube: unpressurized from $R_i = 3.0$ to $R_o = 3.6\,\text{mm}$; under pressure the inner surface reaches $r_i = 3.3\,\text{mm}$. Then $r = \sqrt{1.89 + R^2}$ (because $3.3^2 - 3.0^2 = 1.89$) and $r_o = \sqrt{14.85} = 3.854\,\text{mm}$. The inner surface stretches $3.3/3.0 = 1.100$ and the outer one $3.854/3.6 = 1.070$ (Fig. 9): **in a thick wall there is no single stretch**.

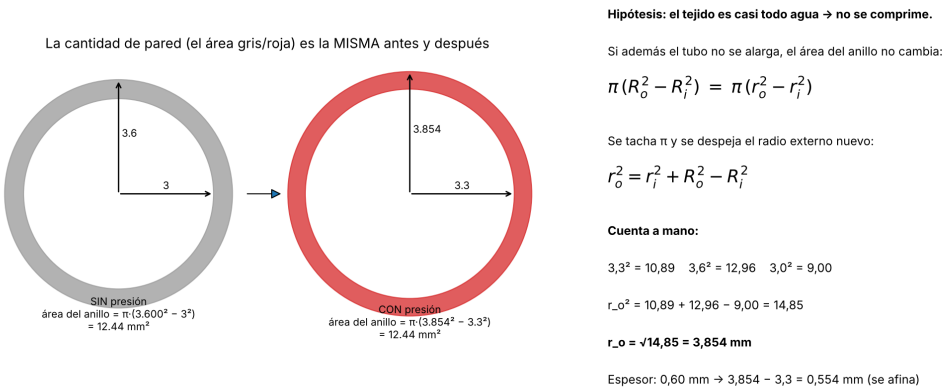
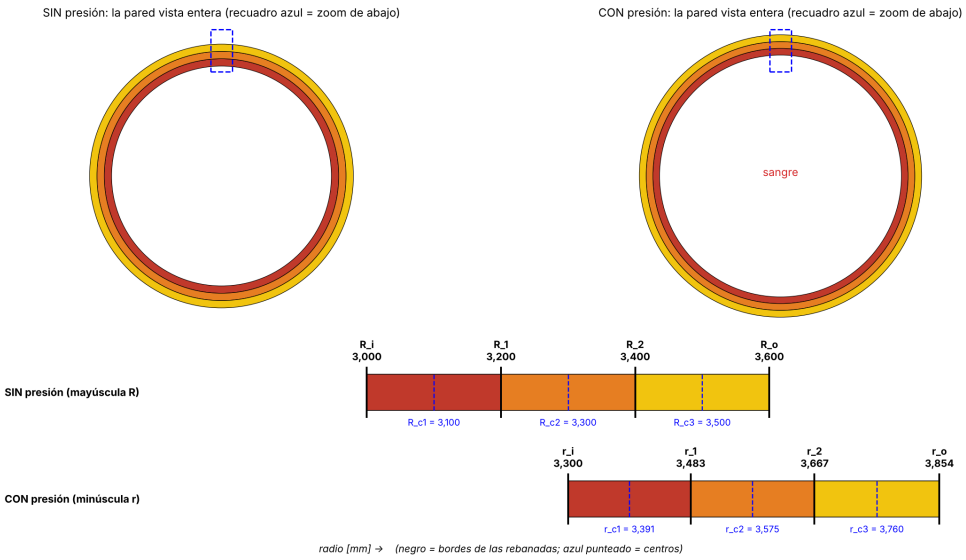


Figure 16: The ring area (gray unpressurized, red pressurized) is the same: $\pi(3.6^2 - 3.0^2) = \pi(3.854^2 - 3.3^2) = 12.44\,\text{mm}^2$.

B.3 The thick wall as a stack of thin walls

The wall is cut into thin slices. Each slice is a thin wall, with its own stretch $\lambda = r_c/R_c$ (at its center), and contributes $\sigma\,\Delta r/r_c$ (Laplace); the pressure is the sum, which in the limit is the integral of Eq. (10).



Nombre	Qué es	Sin presión R [mm]	Cuenta: r = √(3,3² + R² − 3,0²) = √(1,89 + R²)	Con presión r [mm]
R_i → r_i	cara interna (borde)	3,0	√(1,89 + 9,00) = √10,89	3,300
R_c1 → r_c1	centro de la rebanada 1	3,1	√(1,89 + 9,61) = √11,50	3,391
R_1 → r_1	borde entre rebanadas 1 y 2	3,2	√(1,89 + 10,24) = √12,13	3,483
R_c2 → r_c2	centro de la rebanada 2	3,3	√(1,89 + 10,89) = √12,78	3,575
R_2 → r_2	borde entre rebanadas 2 y 3	3,4	√(1,89 + 11,56) = √13,45	3,667
R_c3 → r_c3	centro de la rebanada 3	3,5	√(1,89 + 12,25) = √14,14	3,760
R_o → r_o	cara externa (borde)	3,6	√(1,89 + 12,96) = √14,85	3,854

Figure 17: The example tube in 3 slices: boundaries (black) and centers (blue), unpressurized (R) and pressurized ($r = \sqrt{1.89 + R^2}$).

With a material without knot, $\sigma = 1000\,(\lambda - 1)$ kPa:

	Slice 1 (inner)	Slice 2	Slice 3 (outer)
unpressurized center R_c [mm]	3.1	3.3	3.5
pressurized center $r_c = \sqrt{1.89 + R_c^2}$ [mm]	3.391	3.575	3.760
stretch $\lambda = r_c/R_c$	1.094	1.083	1.074
stress σ [kPa]	93.9	83.3	74.4
inflated thickness Δr [mm]	0.183	0.185	0.186
contribution $\sigma\,\Delta r/r_c$ [kPa]	5.06	4.30	3.68

The total pressure is $5.06 + 4.30 + 3.68 = 13.05$ kPa (with thousands of slices, 13.07). The inner slice contributes more because it stretches more.

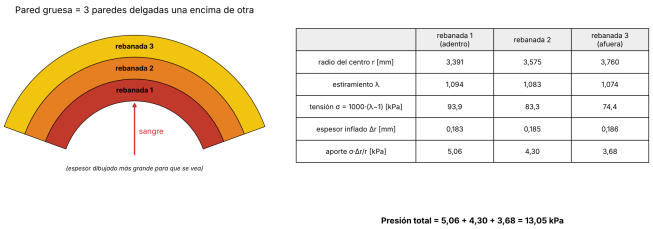


Figure 18: Thick wall = three thin walls, one on top of the other.

B.4 In a thick wall, the position of each layer matters

The wall now has two materials with knots, $\sigma = E\max(0, \lambda - k)$: a blue layer ($k = 1.02$, $E = 400$ kPa) from 3.0 to 3.3 mm and a green one ($k = 1.08$, $E = 1500$ kPa) from 3.3 to 3.6 mm, each

cut into 3 slices.

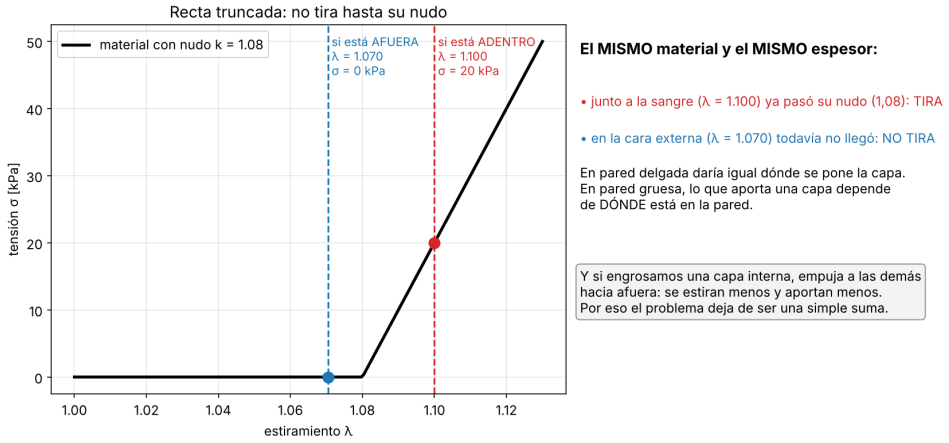


Figure 19: A material with knot 1.08 (in the drawing, $E = 1000\,\text{kPa}$) bears load next to the blood ($\lambda = 1.100$: $\sigma = 1000 \times 0.020 = 20\,\text{kPa}$) and bears no load at the outer surface ($\lambda = 1.070 < 1.08$: $\sigma = 0$). The same material and the same thickness contribute differently depending on where they are.

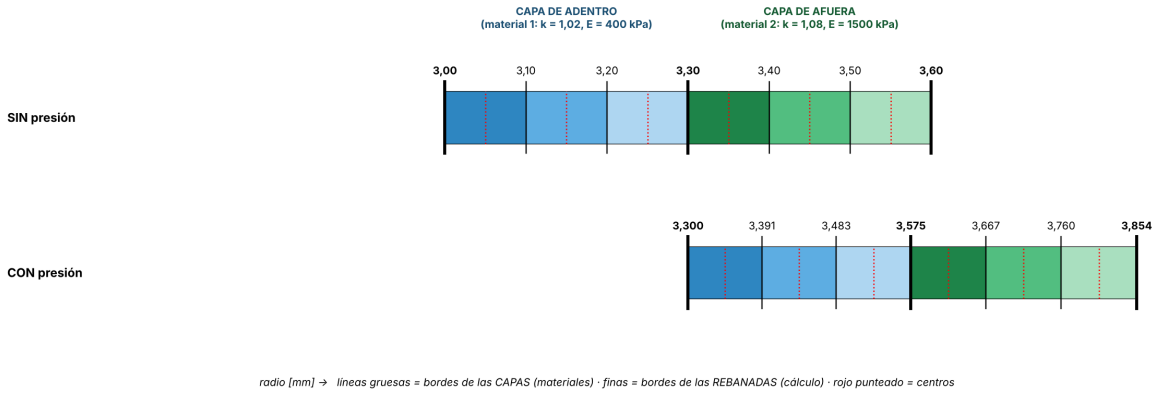


Figure 20: Two 0.3 mm layers, unpressurized (top) and pressurized (bottom). The blue layer contributes $0.84 + 0.76 + 0.69 = 2.28\,\text{kPa}$; of the green one, only the first slice bears load ($\lambda = 1.081$): $0.04\,\text{kPa}$.

If only the thickness of the blue layer is changed, the green one —untouched— changes its contribution, because it is pushed outward and stretches less:

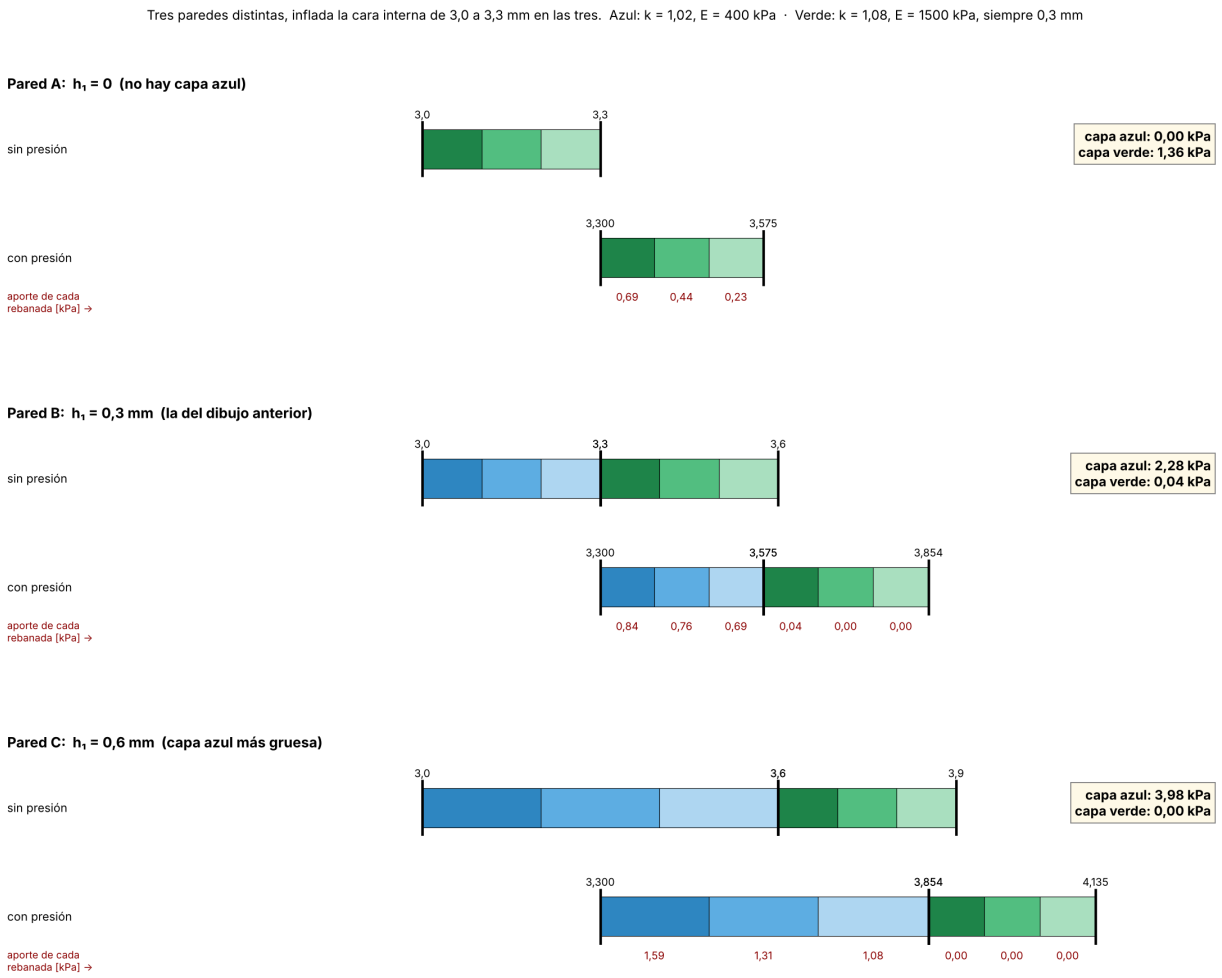


Figure 21: Three walls with the same 0.3 mm green layer and the same inner surface inflated to 3.3 mm. Without a blue layer the green one contributes 1.36 kPa; with 0.3 mm of blue, 0.04; with 0.6 mm, nothing. Doubling the blue layer raises its contribution from 2.28 to 3.98 kPa, which is not twice as much: the problem is nonlinear in the thicknesses.

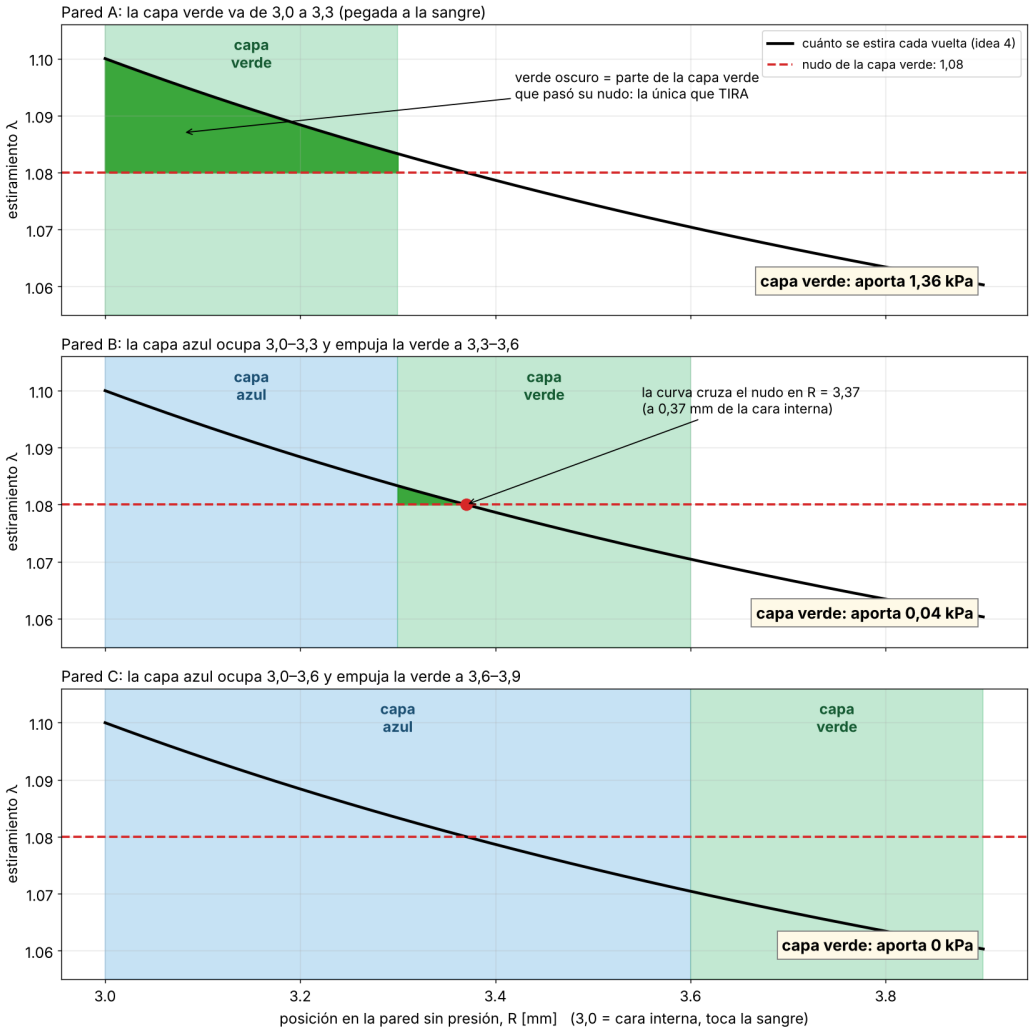


Figure 22: The same three walls seen as the stretch of each circumferential fiber versus its unpressurized radius (black curve) and the knot of the green layer (red line). The green layer contributes only in the part of it lying above 1.08 (dark green): in A, its whole inner part; in B, a sliver next to $R = 3.37\,\text{mm}$; in C, nothing.

B.5 The prestretch δ

δ shifts all the knots together: the blue one moves to $1.02 + \delta$ and the green one to $1.08 + \delta$. The calculation for the two-layer wall (blue from 3.0 to 3.3, green from 3.3 to 3.6 mm, inner surface inflated to 3.3 mm) is repeated with three values of δ. The stretches of the slices do not change (they depend only on the geometry); what changes is where each material starts bearing load.

		λ	σ [kPa]		
Layer	center R [mm]		$\delta = -0.03$	$\delta = 0$	$\delta = +0.03$
blue, $\sigma = 400\max(0, \lambda - 1.02 - \delta)$	3.05	1.097	42.8	30.8	18.8
	3.15	1.091	40.4	28.4	16.4
	3.25	1.086	38.3	26.3	14.3
green, $\sigma = 1500\max(0, \lambda - 1.08 - \delta)$	3.35	1.081	46.4	1.4	0
	3.45	1.076	39.7	0	0
	3.55	1.072	33.6	0	0
contribution of the blue layer [kPa]			3.24	2.28	1.32
contribution of the green layer [kPa]			3.00	0.04	0
total pressure [kPa]			6.25	2.33	1.32

Verification example: first blue slice with $\delta = -0.03$: the knot is $1.02 - 0.03 = 0.99$ and $\sigma = 400 \times (1.097 - 0.99) = 42.8\,\text{kPa}$; its contribution is $42.8 \times 0.091/3.346 = 1.16\,\text{kPa}$. With $\delta = -0.03$ the whole green layer passes its knot (1.05) and contributes as much as the blue one; with $\delta = +0.03$ the green knot rises to 1.11 and the green layer does not work. Thus, δ selects *which materials are active* in the stretch interval of each subject: a soft subject, which stretches a lot, needs knots further to the right ($\delta > 0$), and a stiff one, further to the left ($\delta < 0$). Without δ, each subject would see only one or two of the knots of the basis.

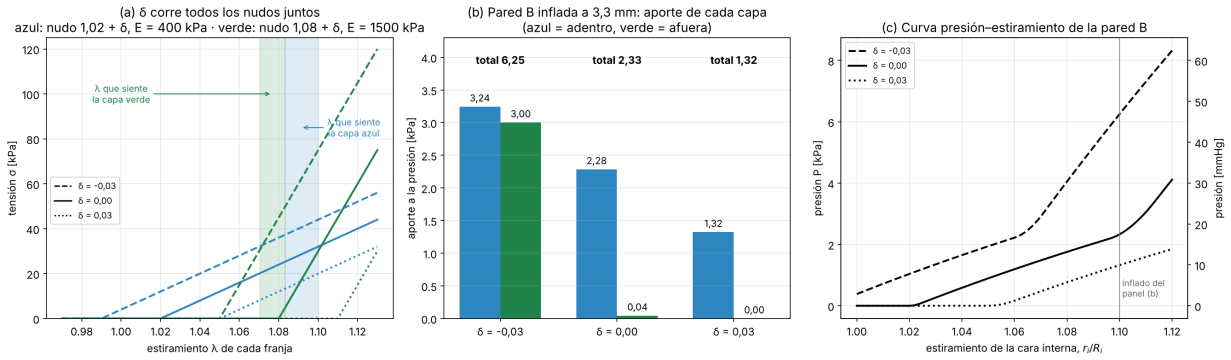


Figure 23: Effect of δ on the two-layer wall. (a) Law of each material for $\delta = -0.03$, 0 and $+0.03$; the shaded bands are the stretches experienced by each layer with the inner surface at 3.3 mm. (b) Contribution of each layer to the pressure in that state. (c) Pressure–stretch curve of the inner surface (right axis in mmHg): δ shifts the point at which each layer engages and changes the shape of the curve. The values belong to a toy example (0.3 mm layers and low moduli), not to a design.

C Training of the neural networks

Four networks were trained: the first-level chained network, which maps the curve $\sigma_{\theta\theta}(\lambda_\theta)$ to the integer laminae of each material (Section 3.5), and the thick-wall network (block A4), each with 6 and with 5 materials. Fig. 24 shows the training and validation loss epoch by epoch, and Table 11 summarizes the training and the final loss on the three data sets. The test loss is computed on curves the network never saw, neither to fit the weights nor to decide when to stop.

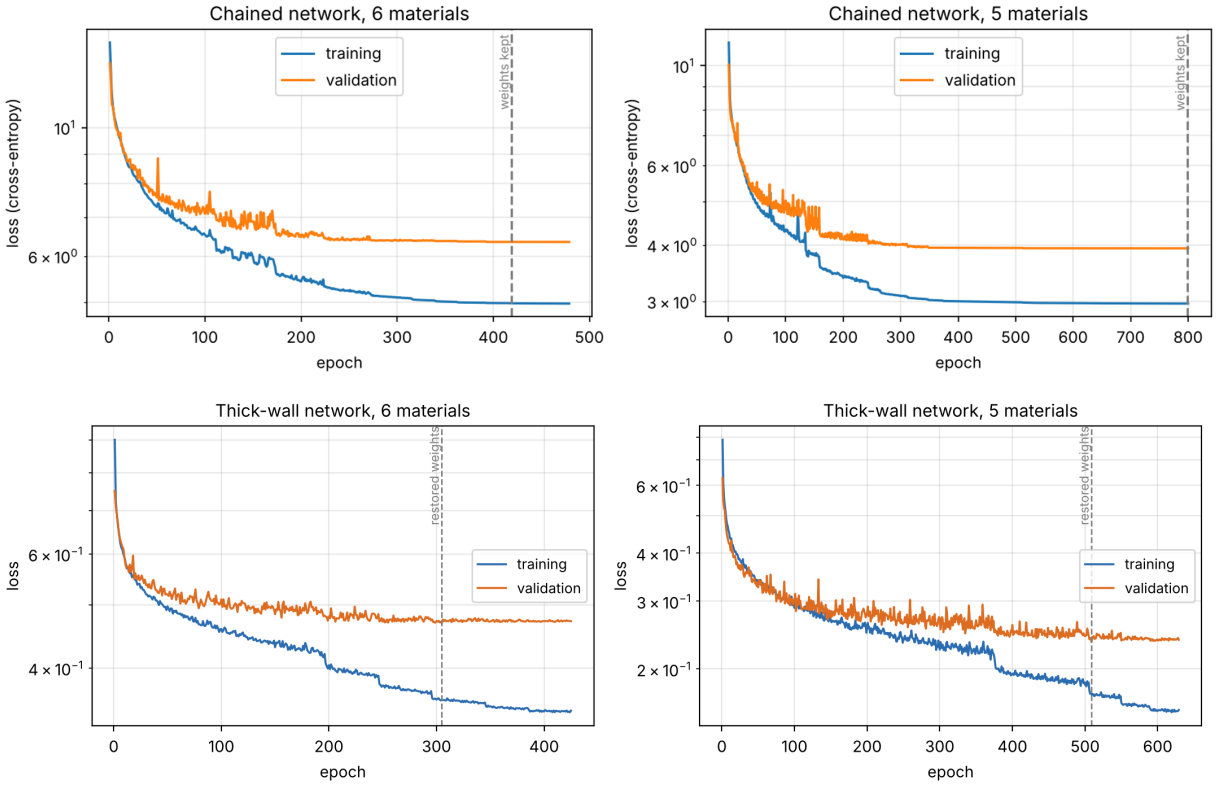


Figure 24: Learning curves: training (blue) and validation (orange) loss per epoch, on a logarithmic scale. The dashed line marks the epoch whose weights are kept. The loss logged during training includes the L2 regularization term. Top: first-level chained network (cross-entropy summed over the materials); bottom: thick-wall network (mean squared error of the standardized design).

Table 11: Training summary. Epochs: total run and epoch whose weights are kept. Loss without the L2 term, computed with the kept weights; the chained network uses the cross-entropy summed over the materials (in nats) and the thick-wall network the mean squared error of the standardized design, so the two sets of values are not comparable with each other. The last column is the test loss divided by the training loss.

Network	Data (train/val/test)	Epochs	LR reductions	Train	Val.	Test	Test/train
Chained network, 6 mat.	24 151 / 3451 / 6901	479 / 419	10	4.96	6.34	6.31	1.27
Chained network, 5 mat.	24 236 / 3463 / 6925	800 / 799	10	2.95	3.92	4.01	1.36
Thick wall, 6 mat.	11 220 / 1530 / 2250	425 / 305	5	0.342	0.462	0.484	1.42
Thick wall, 5 mat.	11 220 / 1530 / 2250	629 / 509	4	0.153	0.227	0.250	1.64

First-level chained network. The validation loss separates from the training loss within the first few tens of epochs and stabilizes, while the training loss keeps decreasing; early stopping keeps the weights of the validation minimum, and the test loss stays next to the validation loss (6.31 versus 6.34 and 4.01 versus 3.92): the network does not degrade on new curves. The 5-material network reached the maximum of 800 epochs with the validation loss already stable. That the

loss does not drop further has a reason of its own: several integer combinations reproduce almost the same curve, and the label (the minimax of the noisy curve) picks only one. The network gives exactly the minimax laminae for 6.1 % of the test curves with 6 materials and 16.7 % with 5, and is within one lamina of it in every material for 31.9 and 58.6 %; even so, its design meets the target for 1988 and 1984 of the 2000 working-space curves (Table 2). The network is therefore judged by the error on the curve, which is the design criterion, and not by matching the minimax.

Thick-wall network. Here validation departs from training within the first 50 epochs and levels off, while the training loss keeps decreasing. Early stopping keeps the weights at the validation minimum, and the test loss stays close to the validation loss (0.484 vs. 0.462 and 0.250 vs. 0.227): the network does not degrade on new data. The loss is nevertheless high, and the reason lies in the problem, not in the network: the thick-wall inverse problem (from the curve to the design) is ill-conditioned.

Conditioning. For 60 arteries from the physiological families, the Jacobian J of the graft curve with respect to the design $\theta = (h_1 \dots h_N, \delta)$ was computed by finite differences, with each parameter measured in standard deviations (the scale used by the network) and the curve in % of its range. Its singular values tell how much the curve moves when the design moves in each direction. With 6 materials, in the most sensitive direction the curve changes by about 850 % of its range per standard deviation (8.5 times the range); in the least sensitive one, by only 1 % (with 5 materials: 750 % and 2.6 %). The condition number $\kappa(J)$ has a median of 900 with 6 materials and 300 with 5 (90th percentile: $2.4 \cdot 10^4$ and $1.5 \cdot 10^4$), compared with $\mathrm{cond}(A) = 28$ at the first level. The least sensitive direction is typically more thickness of one material and less of its neighbor: it changes the design by one standard deviation and the curve by 1 %, the same order as the 0 to 1 % noise used to train the network. From the curve, such designs are indistinguishable for any method, and the network cannot guess which one is the label. This explains the high loss, and why 5 materials, which is better conditioned, has half the loss of 6.

The measure that matters. For the graft, two designs whose curves differ by less than 1 % are equally good; ill-conditioning means that there are many acceptable designs, not that the design is uncertain. The network is therefore judged by the curve error, which is the design criterion. Over 500 test curves, the curve of the design proposed by the network has a median worst-point error of 6.1 % of the range with 6 materials and 4.3 % with 5: 3 to 4 % for arteries and 11 to 12 % for grafts with random designs, which cover the whole design space. That is not enough to design with, and the network is not used that way: it is the starting point of the A2 fit, which brings the design to the target (Section 4).

D Laminae and errors of each curve

Integer laminae of each material and errors of the chained network, of the chained network with the ±1 search and of the minimax, for the 41 Masson subjects and for the 63 external curves inside y_1–y_2, with 6 and with 5 materials. Laminae are listed from material 1 to N.

Table 12: The 41 Masson subjects, 6 materials: integer laminae of each material ($n_1 \dots n_6$) and errors against the subject's curve, E_{rms} and $e_{\max}$ [% of range]. Set: T, training; E, test. In bold, $e_{\max} > 3\,\%$.

Subject	Set	Chained network			Chained network + ±1			Minimax		
		laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
NT01	T	5 7 5 6 6 7	0.66	1.11	5 7 5 6 6 7	0.66	1.11	5 7 5 6 7 5	0.72	1.11
NT02	E	9 3 2 5 3 4	0.54	1.55	9 3 2 5 3 4	0.54	1.55	9 3 3 3 4 5	0.66	1.31
NT03	T	9 4 0 3 1 2	1.11	1.96	9 4 0 3 1 2	1.11	1.96	9 4 0 3 2 0	1.21	1.96
NT04	T	11 2 0 0 2 0	1.41	2.34	11 2 0 0 2 0	1.41	2.34	11 2 0 0 2 0	1.41	2.34
NT05	T	11 5 5 2 4 3	0.62	1.53	11 5 5 2 4 3	0.62	1.53	11 6 1 5 4 1	0.94	1.45
NT06	T	7 6 6 3 5 4	0.62	1.61	7 6 6 4 4 4	0.48	1.51	7 7 3 5 4 4	0.89	1.41
NT07	T	11 6 5 3 3 2	0.77	2.26	10 7 4 3 3 2	1.52	2.18	10 8 2 3 4 1	0.55	1.34
NT08	E	5 3 2 1 2 1	0.81	2.82	5 3 2 2 1 1	0.74	1.61	5 3 2 2 1 1	0.74	1.61
NT09	T	9 6 8 6 5 6	0.74	1.86	9 6 8 6 5 6	0.74	1.86	8 8 5 7 5 6	0.49	0.92
NT10	E	13 11 16 16 12 15	0.56	1.23	13 11 16 15 13 14	0.39	1.01	13 12 13 15 15 12	0.56	0.97
NT11	E	11 14 16 17 15 16	0.68	2.02	12 13 16 17 16 16	0.41	1.02	12 13 16 18 15 16	0.47	1.02
NT12	T	7 6 5 5 5 5	0.36	1.02	7 6 6 4 5 5	0.47	0.96	7 6 6 4 5 5	0.47	0.96
NT13	T	8 7 5 6 4 5	0.77	1.44	8 7 5 5 5 5	0.81	1.44	8 7 5 5 7 1	0.87	1.44
NT14	T	7 6 3 4 4 3	0.62	1.47	7 6 3 4 4 3	0.62	1.47	7 6 3 4 4 3	0.62	1.47
NT15	T	4 6 5 4 4 4	0.56	1.38	4 6 5 4 4 4	0.56	1.38	4 6 5 4 4 5	0.61	1.38
NT16	T	3 12 18 11 15 14	1.02	1.70	2 13 17 11 15 13	0.58	1.16	2 13 15 14 14 13	0.38	0.89
HT01	T	5 9 5 7 7 6	0.46	0.79	5 9 5 7 7 6	0.46	0.79	5 9 5 7 7 6	0.46	0.79
HT02	T	12 5 6 4 4 3	0.74	1.83	12 5 6 4 4 3	0.74	1.83	11 7 3 4 6 1	0.60	1.40
HT03	E	4 9 7 6 7 6	0.36	0.95	4 9 7 6 7 7	0.39	0.84	4 9 7 6 7 7	0.39	0.84
HT04	T	8 3 1 2 1 2	0.86	2.16	8 3 2 1 1 2	0.75	1.59	8 3 2 1 1 2	0.75	1.59
HT05	E	13 12 10 11 9 11	0.60	1.99	13 12 10 12 9 10	0.38	1.07	13 12 10 11 12 7	0.42	1.09
HT06	T	12 14 14 17 15 14	0.64	1.29	12 14 14 16 15 14	0.48	1.03	13 12 18 14 16 14	0.49	0.78
HT07	T	14 9 14 9 9 9	0.74	1.86	13 10 13 8 10 9	0.77	1.75	13 11 11 9 10 8	0.43	0.70
HT08	T	10 12 14 14 13 14	0.75	1.16	9 13 13 14 14 13	0.42	1.04	10 11 17 12 16 10	0.51	1.00
HT09	T	19 11 15 8 14 8	0.73	1.58	20 10 15 9 13 9	0.50	0.91	20 10 14 10 12 12	0.38	0.91
HT10	T	2 10 4 8 5 7	0.49	1.19	2 10 5 7 5 7	0.65	1.13	2 10 5 6 7 5	0.54	0.97
HT11	T	7 5 0 4 2 3	0.87	2.30	7 5 0 4 2 3	0.87	2.30	7 5 1 2 3 3	1.11	2.30
HT12	T	8 6 5 3 4 3	0.39	1.13	8 6 5 4 3 3	0.55	1.09	8 6 5 4 3 3	0.55	1.09
HT13	E	16 4 7 4 6 5	0.51	1.59	16 4 7 4 7 4	0.45	1.22	16 4 7 4 7 4	0.45	1.22
HT14	E	8 6 5 5 4 5	0.43	0.95	8 6 5 5 4 5	0.43	0.95	8 6 5 5 4 5	0.43	0.95
HT15	T	10 10 8 9 7 7	0.39	0.79	10 10 8 9 7 7	0.39	0.79	10 10 8 9 7 7	0.39	0.79
HT16	T	6 15 15 14 15 14	0.34	0.95	6 15 15 14 15 14	0.34	0.95	6 16 12 15 15 14	0.47	0.82
HT17	T	12 13 16 18 13 16	0.78	1.17	11 14 16 17 13 15	0.48	0.98	11 14 16 16 15 15	0.43	0.79
HT18	T	10 5 5 4 5 3	0.38	1.26	10 5 5 4 5 4	0.42	1.05	10 5 5 4 5 4	0.42	1.05
HT19	E	9 6 5 5 3 5	0.41	1.42	9 6 5 5 4 4	0.42	0.97	9 6 6 3 5 4	0.39	0.94
HT20	T	3 10 8 7 7 7	0.48	1.55	3 10 8 7 8 6	0.41	1.01	3 10 8 7 8 7	0.44	1.01
HT21	T	11 5 6 3 4 4	0.51	1.25	11 5 6 3 4 4	0.51	1.25	11 5 5 5 3 4	0.59	1.25
HT22	T	13 10 9 5 6 5	0.37	0.97	13 10 8 6 6 5	0.33	0.97	13 10 9 5 6 5	0.37	0.97
HT23	T	8 7 8 10 9 10	0.90	1.50	7 8 8 9 10 9	0.42	0.97	7 8 8 9 10 9	0.42	0.97
HT24	E	13 8 10 8 8 8	0.36	1.12	13 8 10 8 8 7	0.35	1.12	13 9 7 9 9 6	0.70	1.07
HT25	T	5 16 19 20 19 20	1.00	2.90	6 15 19 19 18 19	0.37	0.88	6 15 19 19 19 17	0.40	0.82

Table 13: The 41 Masson subjects, 5 materials: integer laminae of each material ($n_1 \dots n_5$) and errors against the subject's curve, E_{rms} and $e_{\max}$ [% of range]. Set: T, training; E, test. In bold, $e_{\max} > 3\,\%$.

Subject	Set	Chained network			Chained network + ±1			Minimax		
		laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
NT01	T	5 7 5 7 5	0.72	1.86	5 7 5 7 6	0.77	1.49	5 7 5 7 6	0.77	1.49
NT02	E	9 3 3 4 4	0.88	2.03	9 3 3 4 4	0.88	2.03	9 3 3 4 4	0.88	2.03
NT03	T	9 4 0 3 1	1.57	2.56	9 4 0 3 1	1.57	2.56	9 4 0 3 1	1.57	2.56
NT04	T	11 2 0 1 0	2.32	**3.50**	11 2 0 1 0	2.32	**3.50**	11 2 0 1 0	2.32	**3.50**
NT05	T	11 5 5 2 3	0.78	1.99	11 5 4 3 4	0.86	1.99	11 6 2 4 3	1.18	1.90
NT06	T	7 7 3 5 4	1.00	1.68	7 7 3 5 4	1.00	1.68	7 7 3 5 4	1.00	1.68
NT07	T	11 6 5 2 3	0.81	2.31	11 6 4 3 3	1.17	2.31	10 8 2 4 1	0.87	1.76
NT08	E	5 3 2 1 2	0.57	1.62	5 3 2 1 2	0.57	1.62	5 3 2 1 2	0.57	1.62
NT09	T	8 8 5 6 6	0.68	2.03	8 8 6 5 6	0.69	1.36	8 8 6 5 6	0.69	1.36
NT10	E	12 13 14 14 14	0.63	1.62	13 12 15 14 14	0.72	1.13	13 12 15 14 14	0.72	1.13

Subject	Set	Chained network			Chained network + ±1			Minimax		
		laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
NT11	E	12 13 17 16 16	0.64	1.83	13 12 18 16 16	0.69	1.21	13 12 18 16 16	0.69	1.21
NT12	T	7 6 5 5 5	0.54	1.52	7 6 5 5 5	0.54	1.52	7 6 5 5 5	0.54	1.52
NT13	T	7 8 5 5 4	0.69	2.01	8 7 6 4 5	0.96	1.48	8 7 6 4 5	0.96	1.48
NT14	T	7 6 3 4 4	0.74	1.81	7 6 3 4 4	0.74	1.81	7 6 3 4 4	0.74	1.81
NT15	T	4 6 5 4 4	0.65	1.40	4 6 5 4 4	0.65	1.40	4 6 5 4 4	0.65	1.40
NT16	T	2 13 16 13 14	0.64	1.93	3 12 16 14 13	0.72	1.67	2 14 13 16 12	0.75	1.40
HT01	T	6 7 9 5 7	0.97	2.26	6 7 8 6 7	0.98	2.26	5 9 6 6 8	0.73	1.42
HT02	T	11 7 3 5 3	0.69	1.70	11 7 3 5 3	0.69	1.70	11 7 3 5 3	0.69	1.70
HT03	E	4 9 7 6 7	0.52	1.31	4 9 7 6 7	0.52	1.31	4 9 7 6 7	0.52	1.31
HT04	T	8 3 1 2 1	0.73	2.00	8 3 1 2 1	0.73	2.00	8 3 1 2 1	0.73	2.00
HT05	E	14 11 11 11 10	0.70	1.27	14 11 12 10 9	0.79	1.24	14 11 12 10 9	0.79	1.24
HT06	T	12 14 15 16 14	0.59	1.26	12 14 15 16 14	0.59	1.26	12 14 15 16 14	0.59	1.26
HT07	T	12 13 8 10 9	0.74	1.77	13 12 8 11 8	0.92	1.60	13 11 11 9 10	0.58	1.22
HT08	T	9 13 14 13 14	0.60	1.76	10 12 15 13 13	0.77	1.22	10 12 15 13 13	0.77	1.22
HT09	T	20 11 11 12 10	0.67	1.59	20 11 11 12 11	0.66	1.19	20 11 11 13 9	0.71	1.19
HT10	T	3 8 8 5 7	0.87	2.33	3 8 8 5 7	0.87	2.33	2 10 5 7 6	0.73	1.44
HT11	T	8 3 4 2 3	1.15	2.50	8 3 4 2 2	1.09	2.50	8 3 4 2 2	1.09	2.50
HT12	T	8 6 5 3 3	0.56	1.48	8 6 5 3 3	0.56	1.48	8 6 5 3 3	0.56	1.48
HT13	E	16 4 6 6 4	0.73	2.35	16 4 6 6 5	0.70	1.92	16 5 4 7 4	1.11	1.65
HT14	E	8 6 5 5 4	0.54	1.02	8 6 5 5 4	0.54	1.02	8 6 5 5 4	0.54	1.02
HT15	T	10 10 9 7 8	0.57	1.04	10 10 9 7 8	0.57	1.04	10 10 9 7 8	0.57	1.04
HT16	T	6 16 14 14 15	0.68	1.27	6 16 14 14 15	0.68	1.27	6 16 13 16 13	0.61	1.27
HT17	T	11 15 15 16 15	0.88	1.43	11 15 14 17 14	0.74	1.38	12 13 17 16 14	0.71	1.19
HT18	T	10 5 5 4 4	0.54	1.67	10 5 5 4 5	0.57	1.59	10 5 5 4 5	0.57	1.59
HT19	E	9 6 5 4 4	0.62	1.99	9 6 5 5 3	0.62	1.51	9 6 6 3 5	0.79	1.51
HT20	T	2 12 5 8 7	0.79	2.05	3 11 6 8 6	1.01	1.58	3 11 6 8 6	1.01	1.58
HT21	T	11 5 5 4 3	0.72	1.90	11 5 5 4 4	0.71	1.82	11 5 6 3 4	0.96	1.82
HT22	T	13 10 8 5 6	0.47	1.55	13 10 8 5 5	0.51	1.55	13 11 5 8 3	0.88	1.47
HT23	T	7 8 9 9 10	0.66	1.28	7 8 9 9 10	0.66	1.28	7 8 9 9 10	0.66	1.28
HT24	E	13 9 8 9 7	0.96	1.62	13 9 8 8 8	0.79	1.34	13 9 8 8 8	0.79	1.34
HT25	T	5 17 18 19 18	0.68	1.55	6 16 18 19 19	0.65	1.20	6 16 18 19 19	0.65	1.20

Table 14: External validation, 6 materials: the 63 curves from other authors inside y_1–y_2. Integer laminae of each material and errors E_{rms} and $e_{\max}$ [% of range]. In bold, $e_{\max} > 3\,\%$.

Study · specimen · curve	Chained network			Chained network + ±1			Minimax		
	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
Sommer · CCA wall · I l	10 6 3 2 1 0	0.51	1.07	10 6 3 2 1 0	0.51	1.07	10 6 3 2 0 2	0.39	0.97
Sommer · CCA wall · II l	11 5 8 2 3 1	0.74	2.12	11 5 7 3 2 2	0.80	2.12	11 6 4 4 2 2	0.94	1.68
Sommer · CCA wall · II r	13 4 8 5 5 4	1.00	1.47	13 4 7 6 4 4	0.77	1.39	13 4 7 7 3 5	0.83	1.39
Sommer · CCA wall · IV l	11 3 5 3 3 0	1.64	2.45	11 3 4 3 3 1	1.58	2.45	11 3 5 3 2 1	1.50	2.45
Sommer · CCA wall · V l	11 5 5 2 3 1	0.81	1.96	11 5 4 3 2 1	0.80	1.96	11 5 4 3 3 0	0.67	1.96
Sommer · CCA wall · V r	13 5 8 5 4 1	0.71	2.60	14 4 7 6 5 1	1.00	1.73	14 4 9 3 7 1	1.07	1.73
Sommer · CCA wall · VI l	10 3 7 2 3 2	0.77	1.98	10 3 6 3 3 2	0.59	1.98	10 4 3 4 4 0	1.21	1.93
Sommer · CCA wall · VI r	12 4 7 5 8 5	0.75	1.46	12 4 7 5 7 6	0.73	1.46	12 4 6 7 6 6	0.50	1.04
Sommer · CCA wall · VII r	11 2 5 5 5 4	2.53	**3.62**	11 1 6 6 4 4	0.67	1.41	11 1 7 4 5 4	0.60	1.32
Sommer · CCA wall · VIII r	10 4 4 4 3 1	0.94	2.91	11 3 3 5 4 1	1.70	2.64	11 2 8 2 5 0	0.95	1.84
Sommer · CCA wall · IX r	11 4 5 3 2 1	1.00	1.74	11 4 5 2 3 0	0.97	1.74	11 4 5 2 3 0	0.97	1.74
Sommer · CCA media–intima · II l	17 9 6 7 7 6	0.55	0.92	17 9 6 7 7 6	0.55	0.92	17 9 6 7 6 8	0.51	0.86
Sommer · CCA media–intima · IV l	17 6 9 7 4 5	0.55	1.52	17 6 9 6 5 4	0.73	1.52	17 7 7 6 7 1	0.79	1.29
Sommer · CCA media–intima · V r	17 8 6 7 4 5	0.69	1.37	17 8 6 7 5 4	0.61	1.12	17 8 7 6 5 4	0.66	1.12
Sommer · CCA media–intima · VI r	17 15 3 5 1 4	0.65	1.31	17 15 4 5 0 3	0.54	0.98	17 15 4 4 3 0	0.52	0.98
Sommer · ICA wall · III l	8 5 1 3 1 1	1.37	2.19	8 5 0 4 1 0	1.29	2.19	8 4 4 1 3 0	1.05	2.00
Sommer · ICA wall · IV l	11 5 4 6 3 1	0.89	2.19	11 5 4 6 3 2	0.84	1.51	11 5 4 7 1 3	0.95	1.51
Sommer · ICA wall · V r	10 4 6 2 4 1	0.77	1.63	10 4 6 2 4 1	0.77	1.63	10 4 6 2 4 1	0.77	1.63
Sommer · ICA wall · VI l	8 4 3 2 1 2	1.17	2.76	8 4 3 2 2 1	0.86	2.48	8 4 3 3 1 0	0.90	2.48
Sommer · ICA wall · VI r	12 5 4 4 1 2	0.65	1.20	12 5 4 4 1 2	0.65	1.20	12 5 4 4 2 0	0.64	1.20
Sommer · ICA wall · VII r	8 4 3 2 1 2	1.66	**3.45**	8 4 3 2 2 1	1.30	2.52	8 4 4 1 3 0	1.09	2.52
Sommer · ICA wall · VIII r	10 3 7 3 3 3	1.08	2.53	11 2 6 4 4 2	1.06	2.00	11 2 7 3 5 0	1.34	2.00
Sommer · ICA wall · IX r	9 3 4 3 1 3	1.57	**3.17**	9 3 3 3 2 2	1.52	2.35	9 3 3 5 0 2	1.59	2.35
Sommer · ICA wall · X r	10 2 7 2 2 2	1.27	2.43	10 2 6 3 2 1	1.36	2.43	9 4 3 4 1 2	0.89	1.76
Sommer · ICA wall · XI r	11 4 5 3 2 1	0.56	1.23	11 4 5 3 2 1	0.56	1.23	11 4 5 3 2 1	0.56	1.23
Sommer · ICA media–intima · IX r	17 8 9 4 4 2	0.58	1.09	17 8 8 5 4 2	0.52	1.09	17 8 8 5 5 0	0.53	1.09
Sommer · ICA media–intima · X r	14 4 10 3 4 1	1.28	**4.22**	15 3 10 4 4 2	1.04	1.91	14 5 7 5 4 2	0.59	1.04
Sommer · ICA media–intima · XI r	14 5 8 3 4 0	0.76	1.90	14 5 8 4 3 0	0.61	1.52	14 5 9 3 3 0	0.72	1.52
Liu · P2	13 2 3 1 0 1	1.02	2.57	13 2 3 0 1 1	1.37	2.57	13 2 4 0 0 1	1.45	2.57
Liu · P4	18 2 0 1 0 0	1.03	1.65	18 2 0 1 0 0	1.03	1.65	18 2 0 1 0 0	1.03	1.65

Study · specimen · curve	Chained network			Chained network + ±1			Minimax		
	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
Liu · P5	11 1 0 0 0 1	2.72	**4.41**	11 1 0 0 1 0	2.37	**3.74**	11 1 0 0 1 0	2.37	**3.74**
Liu · P6	14 1 0 1 0 0	3.65	**4.73**	13 2 0 0 0 0	1.66	**3.53**	13 2 0 0 0 0	1.66	**3.53**
Liu · P7	11 2 0 0 0 0	3.00	**6.24**	12 1 0 0 1 0	1.61	**3.54**	12 1 0 0 1 0	1.61	**3.54**
Liu · P8	12 5 0 1 0 0	1.24	1.94	12 5 0 0 1 0	1.17	1.94	12 5 0 0 1 0	1.17	1.94
Liu · P9	15 6 1 1 0 0	0.72	1.34	15 6 1 0 1 0	0.60	1.34	15 6 1 1 0 0	0.72	1.34
Liu · P10	16 5 0 1 0 0	1.01	1.97	16 5 0 1 0 0	1.01	1.97	16 5 0 1 0 0	1.01	1.97
Kamenskiy · CCA · pt. 1	15 10 4 1 0 0	1.70	2.87	15 11 3 0 1 0	0.37	0.97	15 11 2 1 1 0	0.39	0.97
Kamenskiy · ECA · pt. 1	9 2 6 2 2 1	1.16	1.71	9 2 5 3 2 1	0.77	1.55	9 2 6 1 3 1	0.99	1.55
Kamenskiy · ICA · pt. 1	14 6 4 2 1 1	0.81	1.05	14 6 3 3 1 0	0.53	1.04	14 6 4 1 2 1	0.66	1.04
Kamenskiy · CCA · pt. 4	9 5 0 2 0 0	0.79	2.33	9 5 0 1 1 0	1.21	2.33	9 5 1 0 1 0	0.94	2.33
Kamenskiy · CCA · pt. 5	13 3 1 0 0 0	1.28	2.57	13 3 1 0 0 0	1.28	2.57	13 3 1 0 0 0	1.28	2.57
Kamenskiy · ECA · pt. 5	8 4 3 0 1 0	1.93	**3.75**	8 4 3 0 0 1	2.16	**3.75**	8 5 1 0 1 0	0.96	1.57
Kamenskiy · CCA · pt. 6	7 3 0 2 1 0	2.10	**4.79**	7 3 0 1 2 1	2.67	**4.79**	8 1 3 2 1 0	1.38	2.57
Kamenskiy · ECA · pt. 6	8 3 0 2 0 1	2.06	**3.72**	8 3 0 1 1 0	2.04	**3.72**	9 1 3 1 1 0	1.73	**3.46**
Kamenskiy · ECA · pt. 7	12 1 3 0 1 0	0.89	1.74	12 1 3 0 1 0	0.89	1.74	12 1 3 0 1 0	0.89	1.74
Kamenskiy · ICA · pt. 7	13 4 1 1 1 0	0.81	2.12	13 4 1 1 1 0	0.81	2.12	13 4 1 1 1 0	0.81	2.12
Kamenskiy · CCA · pt. 8	13 0 4 1 2 0	4.10	**7.71**	13 1 3 1 2 0	1.22	2.12	13 1 2 3 1 0	1.08	1.97
Kamenskiy · ECA · pt. 8	15 2 3 0 1 0	0.66	1.15	15 2 3 0 1 0	0.66	1.15	15 2 3 0 1 0	0.66	1.15
Kamenskiy · ICA · pt. 8	14 2 4 1 2 0	0.94	1.80	14 2 4 1 2 0	0.94	1.80	14 2 4 1 2 0	0.94	1.80
Kamenskiy · ECA · pt. 9	9 5 0 2 0 0	1.22	2.38	9 5 0 1 1 0	0.87	2.38	9 5 0 1 1 0	0.87	2.38
Kamenskiy · ICA · pt. 9	17 1 3 1 1 1	2.28	**4.21**	17 1 4 1 0 2	0.98	2.13	17 1 4 1 0 2	0.98	2.13
Kamenskiy · ICA · pt. 10	10 3 2 0 0 0	2.26	**4.42**	10 3 2 0 0 0	2.26	**4.42**	10 4 0 0 0 0	1.33	1.79
Kamenskiy · ICA · pt. 11	12 1 1 0 0 0	4.59	**6.16**	12 2 0 0 0 0	2.98	**5.71**	12 2 0 0 0 0	2.98	**5.71**
Kamenskiy · CCA · pt. 12	9 2 3 0 0 1	1.27	**3.21**	9 2 3 0 1 0	0.70	2.17	9 2 3 0 1 0	0.70	2.17
Kamenskiy · ECA · pt. 12	13 4 0 2 0 0	1.74	2.77	13 4 0 1 1 1	2.06	2.77	13 3 3 1 0 0	1.21	2.13
Kamenskiy · ICA · pt. 12	14 3 0 1 0 1	1.87	**3.71**	14 3 1 0 1 0	0.87	1.34	14 3 1 0 1 0	0.87	1.34
Kamenskiy · ICA · pt. 13	13 2 2 1 1 0	1.05	2.25	13 2 2 1 0 1	0.88	2.25	13 2 2 1 1 0	1.05	2.25
Kamenskiy · CCA · pt. 14	12 1 3 0 1 0	0.75	2.17	12 1 3 0 1 0	0.75	2.17	12 1 3 0 1 0	0.75	2.17
Kamenskiy · ICA · pt. 14	12 0 3 0 1 0	2.34	**3.23**	12 0 2 1 1 0	1.62	2.92	12 0 2 1 1 0	1.62	2.92
Kamenskiy · CCA · pt. 15	12 1 0 0 0 0	4.51	**5.53**	13 0 0 1 0 0	2.37	**4.35**	13 0 0 1 0 0	2.37	**4.35**
Kamenskiy · CCA · pt. 16	13 0 4 2 1 1	2.78	**5.53**	13 1 3 1 2 1	1.80	**3.08**	13 1 2 2 2 1	1.43	2.28
Kamenskiy · ECA · pt. 16	10 2 3 1 1 0	1.01	1.91	10 2 3 1 1 0	1.01	1.91	10 2 3 1 1 0	1.01	1.91
Kamenskiy · ICA · pt. 16	13 4 1 0 0 0	1.36	2.26	13 4 1 0 0 0	1.36	2.26	13 4 1 0 0 0	1.36	2.26

Table 15: External validation, 5 materials: the 63 curves from other authors inside y_1–y_2. Integer laminae of each material and errors E_{rms} and $e_{\max}$ [% of range]. In bold, $e_{\max} > 3\,\%$.

Study · specimen · curve	Chained network			Chained network + ±1			Minimax		
	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
Sommer · CCA wall · I l	10 6 3 1 0	0.73	1.51	10 6 3 1 0	0.73	1.51	10 6 3 1 0	0.73	1.51
Sommer · CCA wall · II l	10 7 4 3 1	0.93	**3.01**	11 6 4 3 2	1.00	1.87	11 6 5 2 2	1.23	1.87
Sommer · CCA wall · II r	12 5 7 5 4	0.95	2.71	13 4 7 6 3	0.86	1.55	13 4 8 4 5	0.97	1.55
Sommer · CCA wall · IV l	10 4 5 2 1	1.53	**3.02**	11 3 4 3 2	1.46	2.63	11 3 5 2 1	1.57	2.63
Sommer · CCA wall · V l	11 5 5 2 0	1.38	2.66	11 5 4 3 0	0.76	2.11	11 5 5 1 2	1.01	2.11
Sommer · CCA wall · V r	13 5 8 3 4	0.93	2.27	14 4 8 4 3	1.03	1.85	14 4 8 4 4	0.99	1.85
Sommer · CCA wall · VI l	10 3 6 2 2	1.07	**3.24**	10 3 5 3 2	1.64	**3.24**	10 4 3 4 2	1.22	2.23
Sommer · CCA wall · VI r	12 4 7 6 6	0.72	1.33	12 4 7 6 6	0.72	1.33	12 4 7 6 6	0.72	1.33
Sommer · CCA wall · VII r	10 3 4 5 5	1.13	2.98	11 2 4 6 4	1.57	2.67	11 2 4 7 2	1.79	2.67
Sommer · CCA wall · VIII r	10 3 6 3 2	2.13	**4.42**	10 4 5 2 3	1.22	2.76	10 4 5 2 4	1.22	2.76
Sommer · CCA wall · IX r	10 5 5 1 2	1.26	**3.19**	11 4 5 1 2	1.11	1.87	11 4 5 1 2	1.11	1.87
Sommer · CCA media–intima · II l	17 9 6 7 7	0.61	1.01	17 9 6 7 7	0.61	1.01	17 9 6 7 7	0.61	1.01
Sommer · CCA media–intima · IV l	17 7 6 7 3	0.94	2.19	17 7 7 6 4	0.80	1.49	17 7 7 6 4	0.80	1.49
Sommer · CCA media–intima · V r	16 9 6 5 5	0.80	1.99	17 8 7 5 5	0.73	1.26	17 8 7 5 5	0.73	1.26
Sommer · CCA media–intima · VI r	17 14 6 2 2	0.49	1.42	17 14 6 1 3	0.51	1.42	17 15 3 3 2	0.72	1.20
Sommer · ICA wall · III l	8 4 3 2 0	1.39	**3.02**	8 4 3 2 1	1.29	2.54	8 4 4 1 0	1.30	2.54
Sommer · ICA wall · IV l	11 4 8 2 3	1.08	**3.22**	12 3 8 2 4	1.35	2.58	11 5 5 4 3	1.00	1.79
Sommer · ICA wall · V r	10 4 5 3 2	0.64	1.73	10 4 5 3 2	0.64	1.73	10 4 5 3 2	0.64	1.73
Sommer · ICA wall · VI l	8 4 3 2 0	0.91	2.69	8 4 3 2 0	0.91	2.69	8 4 4 0 2	1.35	2.69
Sommer · ICA wall · VI r	12 5 4 3 0	0.85	2.63	12 5 4 3 1	0.74	1.44	12 5 4 3 1	0.74	1.44
Sommer · ICA wall · VII r	8 4 3 2 0	1.15	2.74	9 3 3 2 1	1.82	2.69	9 3 3 3 0	1.68	2.69
Sommer · ICA wall · VIII r	10 3 6 3 3	1.04	2.57	11 2 6 4 2	1.17	2.14	11 2 7 2 4	1.33	2.14
Sommer · ICA wall · IX r	9 3 4 2 1	1.48	2.51	9 3 4 2 1	1.48	2.51	9 3 4 2 1	1.48	2.51
Sommer · ICA wall · X r	9 4 4 2 2	1.32	2.05	9 4 3 3 2	1.02	1.86	9 4 3 4 0	1.12	1.86
Sommer · ICA wall · XI r	11 4 5 2 1	0.76	1.44	11 4 5 2 1	0.76	1.44	11 4 5 2 1	0.76	1.44
Sommer · ICA media–intima · IX r	17 8 8 4 3	0.71	1.25	17 8 8 4 2	0.64	1.25	17 8 8 4 3	0.71	1.25
Sommer · ICA media–intima · X r	14 5 7 4 3	0.70	1.31	14 5 7 4 3	0.70	1.31	14 5 7 4 3	0.70	1.31
Sommer · ICA media–intima · XI r	13 7 5 3 2	0.85	2.66	14 6 5 3 3	1.31	2.30	14 6 5 4 2	1.20	2.30

Study · specimen · curve	Chained network			Chained network + ±1			Minimax		
	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$	laminae	E_{rms}	$e_{\max}$
Liu · P2	12 3 2 0 1	2.11	**3.35**	13 2 2 1 0	1.51	2.68	13 2 2 1 0	1.51	2.68
Liu · P4	18 1 1 0 0	4.12	**8.69**	18 2 0 0 1	1.65	2.45	18 2 0 0 1	1.65	2.45
Liu · P5	11 1 0 0 0	2.96	**4.16**	11 1 0 0 1	2.83	**3.91**	11 1 0 0 1	2.83	**3.91**
Liu · P6	14 1 0 1 0	4.51	**5.61**	13 2 0 0 0	2.68	**3.75**	13 2 0 0 0	2.68	**3.75**
Liu · P7	12 1 0 0 0	1.67	**3.16**	12 1 0 0 0	1.67	**3.16**	12 1 0 0 0	1.67	**3.16**
Liu · P8	12 5 0 0 1	1.78	2.45	12 5 0 0 0	1.75	2.45	12 5 0 0 1	1.78	2.45
Liu · P9	15 6 0 1 0	0.80	1.44	15 6 0 1 0	0.80	1.44	15 6 0 1 0	0.80	1.44
Liu · P10	16 5 0 1 0	1.86	2.86	16 5 0 0 1	1.51	2.17	16 5 0 0 1	1.51	2.17
Kamenskiy · CCA · pt. 1	16 10 2 1 1	1.48	2.62	15 11 1 1 0	0.60	1.52	15 11 1 1 1	0.51	1.52
Kamenskiy · ECA · pt. 1	9 3 4 2 1	3.66	**4.63**	9 2 5 2 1	1.04	2.63	9 2 6 0 3	1.42	2.63
Kamenskiy · ICA · pt. 1	14 6 3 2 0	0.72	1.33	14 6 3 2 0	0.72	1.33	14 6 3 2 0	0.72	1.33
Kamenskiy · CCA · pt. 4	9 5 0 1 0	1.24	2.72	9 5 0 1 0	1.24	2.72	9 5 0 1 1	1.18	2.72
Kamenskiy · CCA · pt. 5	13 3 0 0 0	2.46	**5.06**	13 3 0 1 0	1.52	**3.04**	13 3 0 1 0	1.52	**3.04**
Kamenskiy · ECA · pt. 5	8 5 0 1 0	1.11	1.90	8 5 0 1 0	1.11	1.90	8 5 0 1 0	1.11	1.90
Kamenskiy · CCA · pt. 6	7 3 1 2 0	3.79	**5.83**	7 2 2 2 0	2.86	**4.82**	8 1 3 1 1	1.56	2.69
Kamenskiy · ECA · pt. 6	8 3 0 2 0	2.78	**3.98**	8 3 0 1 1	2.46	**3.98**	9 1 3 0 1	1.91	**3.56**
Kamenskiy · ECA · pt. 7	12 1 2 0 1	1.97	**4.11**	12 1 2 1 0	1.39	2.40	12 1 2 1 0	1.39	2.40
Kamenskiy · ICA · pt. 7	13 4 0 2 0	1.43	**3.14**	13 4 1 1 0	1.06	2.29	13 4 1 1 0	1.06	2.29
Kamenskiy · CCA · pt. 8	13 1 3 1 1	1.64	2.46	13 1 2 2 1	1.17	2.19	13 1 2 2 1	1.17	2.19
Kamenskiy · ECA · pt. 8	15 2 2 0 1	0.73	1.92	15 2 2 0 1	0.73	1.92	15 2 2 0 1	0.73	1.92
Kamenskiy · ICA · pt. 8	13 3 3 1 0	1.62	**3.67**	14 2 3 2 0	1.20	2.01	14 2 3 2 0	1.20	2.01
Kamenskiy · ECA · pt. 9	10 3 3 0 0	1.54	**3.66**	10 3 2 1 0	1.90	**3.66**	9 5 0 1 0	1.25	2.78
Kamenskiy · ICA · pt. 9	17 1 3 1 0	1.40	**3.14**	17 1 3 1 1	1.24	2.88	17 1 4 0 0	1.35	2.88
Kamenskiy · ICA · pt. 10	10 3 1 0 0	2.54	**4.38**	10 3 1 0 0	2.54	**4.38**	10 3 1 0 0	2.54	**4.38**
Kamenskiy · ICA · pt. 11	13 1 0 0 0	5.13	**7.29**	12 1 1 0 0	3.55	**5.98**	12 1 1 0 0	3.55	**5.98**
Kamenskiy · CCA · pt. 12	9 2 3 0 0	2.22	**3.97**	9 2 2 1 0	1.37	2.97	9 2 2 1 0	1.37	2.97
Kamenskiy · ECA · pt. 12	13 3 2 0 1	2.48	**4.74**	13 3 3 0 0	1.14	2.77	13 3 3 0 0	1.14	2.77
Kamenskiy · ICA · pt. 12	14 3 0 1 0	1.34	2.09	14 3 1 0 0	1.38	1.78	14 3 1 0 0	1.38	1.78
Kamenskiy · ICA · pt. 13	13 2 2 0 1	0.81	2.23	13 2 2 0 1	0.81	2.23	13 2 2 0 1	0.81	2.23
Kamenskiy · CCA · pt. 14	12 2 0 2 0	3.67	**6.12**	12 2 0 1 0	2.95	**4.80**	12 1 2 1 0	1.25	2.86
Kamenskiy · ICA · pt. 14	11 1 2 0 1	2.32	**5.52**	12 0 2 1 0	2.09	**3.29**	12 0 2 1 0	2.09	**3.29**
Kamenskiy · CCA · pt. 15	13 0 1 0 0	3.14	**6.80**	13 0 0 1 0	2.59	**5.21**	13 0 0 1 0	2.59	**5.21**
Kamenskiy · CCA · pt. 16	13 1 3 2 1	3.16	**6.17**	13 1 2 2 1	1.66	2.76	13 1 2 2 1	1.66	2.76
Kamenskiy · ECA · pt. 16	10 2 3 0 1	1.26	2.03	10 2 3 0 1	1.26	2.03	10 2 3 0 1	1.26	2.03
Kamenskiy · ICA · pt. 16	13 4 0 0 0	2.21	**4.63**	13 4 0 1 0	1.33	2.72	13 4 0 1 0	1.33	2.72

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